AK7455
Zero Latency Angle Sensor IC

1. General Description

The AK7455 is a Silicon monolithic Hall-Effect sensor IC that specializes in detecting rotation angle. A contactless absolute magnetic rotary encoder is easily designed with the AK7455 and a magnet.

The AK7455 detects the horizontal magnetic field vector (Bx, By) with respect to the IC package This is obtained through a magnetic concentrator which is mounted on the Hall-Effect elements. It is advantageous to accurate angular measurements against mechanical misalignment.
It is also possible to set the magnetic field vector from horizontal to vertical (Bx/Bz or By/Bz), which contributes to improving the degree of freedom of positioning of the magnet and IC.

The AK7455 has built-in EEPROMs for angle INL error calibration function. It is possible to reduce the error caused by mounting misalignment etc. by writing the calibration value obtained by external calculation into them.
By using this calibration function, AK7455 can be used not only in the Shaft-End configuration (end of shaft) but also in the Off-Axis configuration (side of shaft).

The AK7455 is the zero-latency rotary angle sensor to follow rotation speed up to 25000rpm by Type2 tracking servo loop architecture. It is suitable for motor-controlled applications with an encoder.
Also, the servo filter bandwidth can be extended to improve the tracking performance of acceleration and deceleration operations.

The AK7455 is equipped with an anomaly magnetic field detection function that judges the magnetic field from sources other than the sensor magnet and issues an alarm using the ERROR pin and register.
The AK7455 also has a function to dynamically reduce angular errors caused by anomaly magnetic fields.

Note1: This datasheet does not describe how to set up the angle INL error calibration, the anomaly magnetic field detection function, or the error reduction function due to anomaly magnetic fields. Please contact us for more information.

2. Applications

Motor Controlled Applications (Robot, Machine tools, Stepping motor, DC brushless motor, etc.)
Optical Encoder Replacement

3. Features

- 360º Contactless Absolute Angle Sensor
- Shaft-End (end of the shaft) and Off-Axis (side of the shaft) Configuration Available
- Operating Temperature: -40 to +125ºC
- Supply Voltage: 3.0 to 5.5V
- Magnetic Field Range: 30 to 70mT (Shaft-End), 10 to 70mT (Off-Axis)
- Angle Resolution: 14bit
- Maximum Tracking Rotation Speed: 25000rpm
- Angle INL: ±0.5º (25ºC, w/o calibration, Shaft-End), ±0.1º (after calibration, typical value)
- Output Delay: 1.2μs (ABZ Hysteresis “OFF” setting)
- Self-diagnostic Functions

- Low Magnetic Flux Density State Detection
- Tracking Lost State Detection
- Anomaly magnetic field detection

☐ Interfaces
- 4-wire Serial Interface (SPI): Absolute Angle Data & User Programming
- ABZ Incremental Output
- UVW Commutation Output

☐ User Programming Functions
- Independently Settable Zero Point for ABZ (linked SPI absolute angle data) and UVW
- Rotation Direction Setting: CCW / CW
- ABZ Incremental Output Resolution Setting: 1 to 4096ppr
- Z-phase Pulse Phase Setting: 4 Edges Selectable
- Z-phase Pulse Width Setting: 1 to 16384LSB
- UVW Commutation Output Resolution Setting: 2 to 64poles (1 to 32pole pairs)
- Band Width Setting of Type2 Tracking Servo Loop Filter
- Angle INL Error Calibration
- Anomaly magnetic field detection
- Function to reduce angular error due to anomaly magnetic field
- X-Y, X-Z and Y-Z magnetic field detection

☐ Package
- 24pin - QFN 4.0mm x 4.0mm x 0.85mm (typ)

☐ Environmental Friendly (RoHS Compliant)
- Lead-free
- Halogen-free

4. Table of Contents

5. Block Diagram and Functional Descriptions

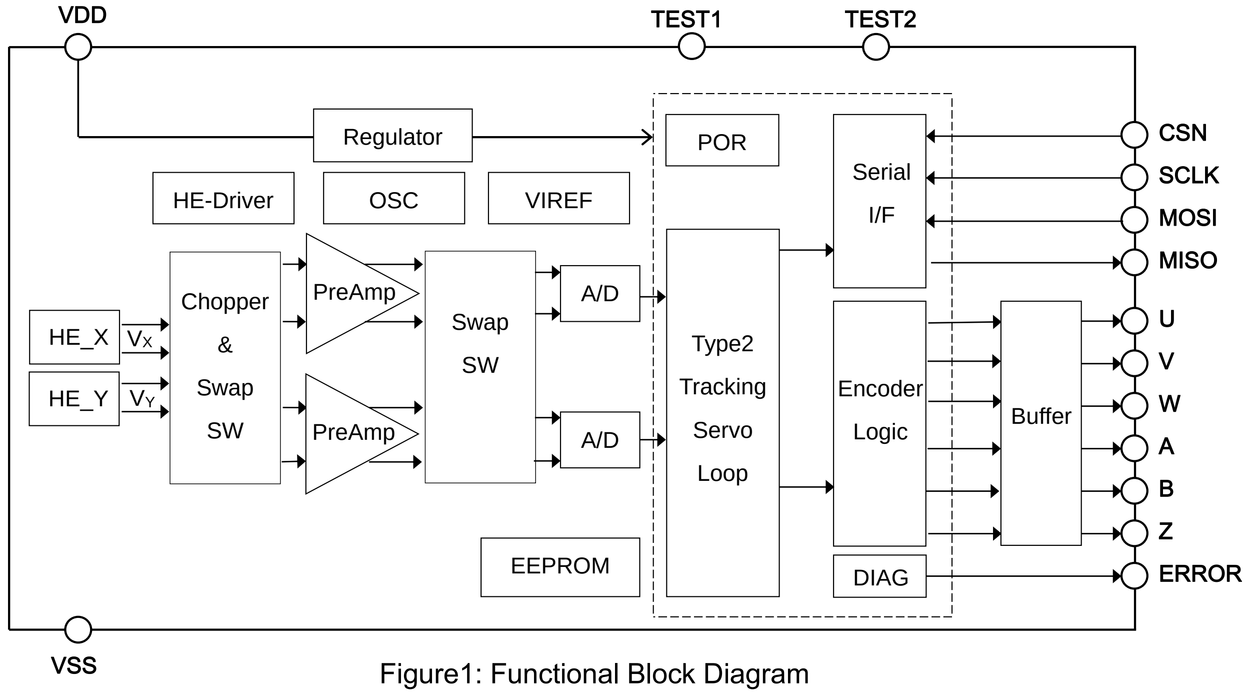


Figure1: Functional Block Diagram

Block name	Functions
HE_X / HE_Y	The magnetic concentrator detects the magnetic field vector components in any two directions and converts them into electrical signals.
Chopper & Swap SW	Switch the current direction to cancel the Hall elements offset. Swap operation reduces X/Y mismatch.
Pre Amp	Amplify the signals from the Hall elements.
Swap SW	Switch the signal path to reduce the error caused by the Hall elements sensitivity mismatch.
A/D	Analog-to-digital converter
Type2 Tracking Servo Loop	Angle data converter that follows the digitally converted input signal
Serial I/F	Interface based on 4-wire SPI protocol
Encoder Logic	Generate the ABZ incremental signal and the UVW commutation signal from the angle data.
DIAG	Self-diagnosis circuit
POR	Generate a reset signal at the time of low supply voltage.
Buffer	Buffer the output signals.
Regulator	Generate internal operating voltage.
HE-Driver	Drive the constant current for the Hall elements.
OSC	Generate the master clock for internal blocks.
VIREF	Generate the reference voltage and current for internal blocks.
EEPROM	Non-volatile memory

6. Pin Configuration and Functional Descriptions

No.	Pin Name	I/O	Type	Functional Descriptions
1	W	O	Digital	W-phase Output
2	V	O	Digital	V-phase Output
3	U	O	Digital	U-phase Output
4	MISO	O	Digital	Serial I/F Data Output
5	MOSI	I	Digital	Serial I/F Data Input
6	SCLK	I	Digital	Serial I/F Clock Input
7	CSN	I	Digital	Serial I/F Chip Select Input
8	NC	-	-	No Connection (Note2)
9	NC	-	-	No Connection (Note2)
10	NC	-	-	No Connection (Note2)
11	VDD	-	Power	Power Supply
12	TEST1	I	-	Test Pin (Note3)
13	TEST2	I	-	Test Pin (Note4)
14	VSS	-	Ground	Ground
15	ERROR	O	Digital	Error Output
16	Z	O	Digital	Z-phase Output
17	B	O	Digital	B-phase Output
18	A	O	Digital	A-phase Output
19	NC	-	-	No Connection (Note2)
20	NC	-	-	No Connection (Note2)
21	NC	-	-	No Connection (Note2)
22	NC	-	-	No Connection (Note2)
23	NC	-	-	No Connection (Note2)
24	NC	-	-	No Connection (Note2)
TAB	TAB	-	-	Back Tab (Note5)

Note2: NC pins must be open.
Note3: TEST1 pin must be open.
Note4: TEST2 pin must be connected to VSS.
Note5: Back Tab must be open.

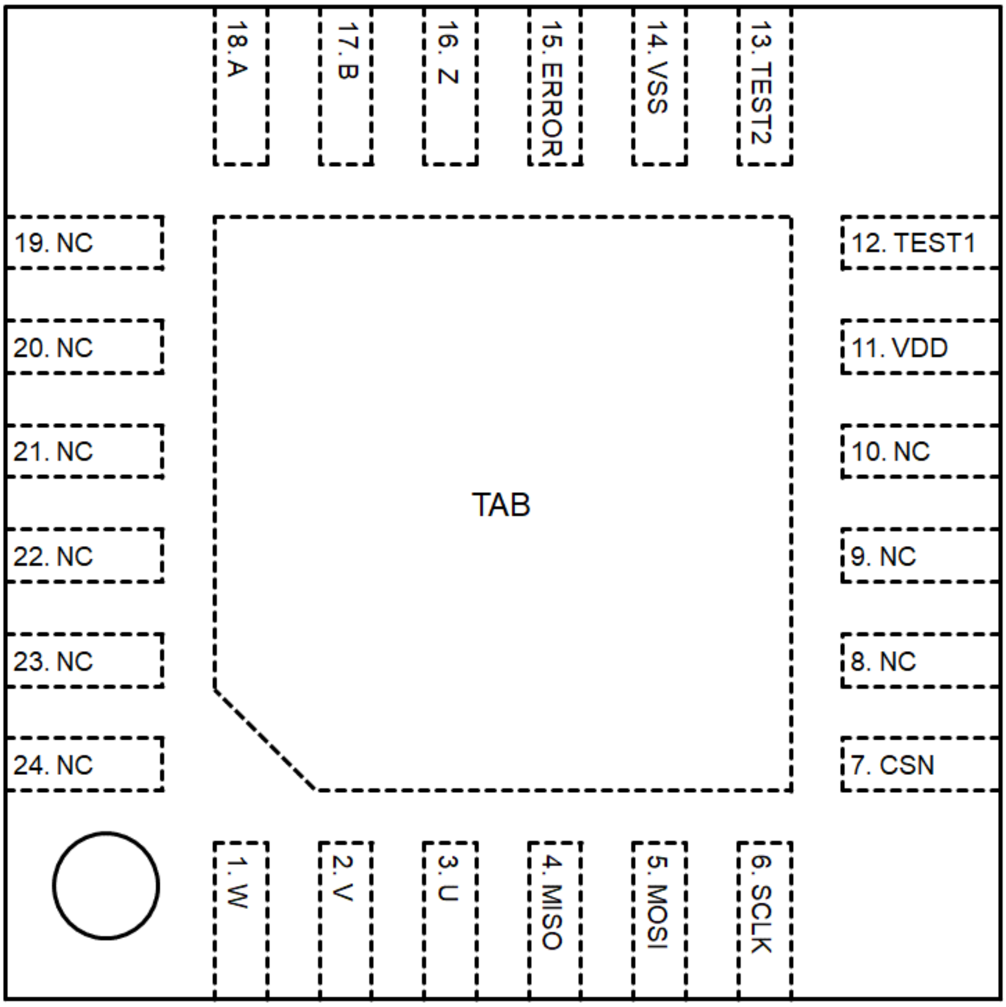


Figure2: Pin Configuration (Top view)

7. Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit	Notes
Supply Voltage	V_{DDA}	-0.3	+6.5	V	VDD pin
Terminal Voltage	V_{TERM}	-0.3	V_{DD}	V	W, V, U, MISO, MOSI, SCLK, CSN, TEST1, TEST2, ERROR, Z, B, A pin
Output Current	I_{OA}	-10	+10	mA	W, V, U, MISO, ERROR, Z, B, A pin
Storage Temperature	T_{STG}	-50	+150	ºC	

WARNING: Operation beyond these limits may cause permanent damage to the device. Even if it does not lead to destruction, reliability and life may be adversely affected. Normal operation is not guaranteed at these extremes. Each voltage is with respect to the VSS pin.

8. Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit	Notes
Supply Voltage	V_{DD}	3.0	5.0	5.5	V	VDD pin
Operating Ambient Temperature	T_a	-40	25	+125	ºC	

WARNING: Electrical and magnetic characteristics are not guaranteed when operated beyond these conditions. Each voltage is with respect to the VSS pin.

9. EEPROM Characteristics

(V_{DD}=3.0 to 5.5V)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Notes
EEPROM Endurance	E_{en}			1000	Cycle	
Ambient Temperature in Writing	T_{aw}	0		+85	ºC	
Writing Time	W_t			5	ms	

10. Electrical and Magnetic Characteristics

(T_a=-40 to +125ºC, V_{DD}=3.0 to 5.5V unless otherwise specified)

Parameter	Symbol	Conditions, Remarks	Min.	Typ.	Max.	Unit
Magnetic Flux Density Range	B_{RG_ON}	Shaft-End configuration	30	50	70	mT
	B_{RG_OFF}	Off-Axis configuration	10		70	mT
Angle Detection Range	A_{RANGE}		0		360	deg.
Angle Resolution	A_{RES}			0.022		deg.
Angle Tracking Rate	T_{RA}	Except for 16-times mode in the filter setting			25000	rpm
	T_{RA16}	16-times mode in the filter setting			10000	rpm
Angle Linearity Error (Note6)	A_{INL_SNC}	Shaft-End configuration, w/o calibration, T_a=25ºC	-0.5		+0.5	deg.
	A_{INL_CAL}	After calibration		±0.1		deg.
Angle Hysteresis (Note6, Note7)	A_{HYS}	T_a=25ºC			0.3	deg.
Thermal Angle Drift (Note6, Note7, Note8)	A_{DRIFT}	T_a=25ºC basis	-0.5		+0.5	deg.
Angle Output Noise (Note6, Note7)	A_{NOISE}	±3.3σ ABZ Hysteresis “1LSB” and Narrow Band Mode Setting	-1		+1	LSB
Angle Output Delay (Note8, Note9)	T_D	ABZ Hysteresis “OFF” setting			1.2	μs
Supply Current	I_{DD}	No output load		15	23	mA
Startup Time (Note10)	T_{START}				40	ms

Note6: When the detection magnetic field direction is set to the X-Y plane.
Note7: This parameter is guaranteed in the magnetic flux density range from 30 to 70 mT.
Note8: This parameter is derived from the IC design and not tested at mass production.
Note9: This parameter changes as follows when the ABZ output hysteresis setting is not OFF.
Output Delay (max) : T_D = 1.2 + 0.125(1 + “ABZ Hysteresis”) [μs]
Note10: Startup time is defined as the time from when VDD reaches the operating voltage level to when the ERROR pin is set to the “H” level.

11. Output Electrical Characteristics

11.1. DC Characteristics

(T_a=-40 to +125ºC, V_{DD}=3.0 to 5.5V unless otherwise specified)

Parameter	Symbol	Conditions, Remarks	Min.	Typ.	Max.	Unit
Output Current	I_{OUT}	W, V, U, ERROR, Z, B, A pin	-2		+2	mA
Output Low Level	V_{OL}	W, V, U, ERROR, Z, B, A pin I_{OUT}=2mA (sink)			0.6	V
		W, V, U, ERROR, Z, B, A pin I_{OUT}=1mA (sink)			0.4	V
Output High Level	V_{OH}	W, V, U, ERROR, Z, B, A pin I_{OUT}=2mA (source)	0.8V_{DD}			V
Output Load Capacitance	C_{OUT}	W, V, U, ERROR, Z, B, A pin			50	pF

11.2. AC Characteristics

(T_a=-40 to +125ºC, V_{DD}=3.0 to 5.5V unless otherwise specified)

Parameter	Symbol	Conditions, Remarks	Min.	Typ.	Max.	Unit
Output Rise Time	T_{RO}	W, V, U, ERROR, Z, B, A pin I_{OUT}=2mA (source) (Time of 0.2V_{DD} to 0.8V_{DD})			30	ns
Output Fall Time	T_{FO}	W, V, U, ERROR, Z, B, A pin I_{OUT}=2mA (sink) (Time of 0.8V_{DD} to 0.2V_{DD})			30	ns

12. Serial I/F Digital Characteristics

12.1. DC Characteristics

(T_a=-40 to +125ºC, V_{DD}=3.0 to 5.5V unless otherwise specified)

Parameter	Symbol	Conditions, Remarks	Min.	Typ.	Max.	Unit
Input Current	I_{SI}	MOSI, SCLK, CSN pin	-10		+10	μA
Input Low Level	V_{LSI}	MOSI, SCLK, CSN pin			0.3V_{DD}	V
Input High Level	V_{HSI}	MOSI, SCLK, CSN pin	0.7V_{DD}			V
Output Current	I_{SO}	MISO pin	-700		+700	μA
Output Low Level	V_{LSO}	MISO pin I_{SO}=700μA (sink)			0.6	V
Output High Level	V_{HSO}	MISO pin I_{SO}=700μA (source)	0.8V_{DD}			V
Output Load Capacitance	C_{SO}	MISO pin			50	pF

12.2. AC Characteristics

(T_a=-40 to +125ºC, V_{DD}=3.0 to 5.5V unless otherwise specified)

Parameter	Symbol	Conditions, Remarks	Min.	Typ.	Max.	Unit
SCLK Frequency	F_{SCLK}		0.001		8000	kHz
	F_{SCLKRE}	@ReadEEPROM OPCODE (Note11)	0.001		100	kHz
SCLK High Time	T_H		50			ns
	T_{HRE}	@ReadEEPROM OPCODE	4			µs
SCLK Low Time	T_L		50			ns
	T_{LRE}	@ReadEEPROM OPCODE	4			µs
Data Output Rise Time	T_{RDO}	Transition time from 0.2V_{DD} to 0.8V_{DD}			20	ns
Data Output Fall Time	T_{FDO}	Transition time from 0.8V_{DD} to 0.2V_{DD}			20	ns
CS Setup Time	T_{CSS}		100			ns
CS Hold Time	T_{CSH}		100			ns
Data Setup Time	T_{DS}		20			ns
Data Hold Time	T_{DH}		20			ns
Data Valid Time	T_{DV}				50	ns
Data Release Time	T_{DR}				100	ns
Wait Time	T_{CW}	@ReadAngle OPCODE	200			ns
	T_{EW}	@WriteEEPROM OPCODE	5			ms
	T_{DW}	@DataRenew OPCODE	150			µs
	T_{RW}	@WriteReg, ReadReg, ReadEEPROM OPCODE	2.5			µs
	T_{MT}	@ChangeMode OPCODE (User Mode to Normal Mode)	40			ms
		@ChangeMode OPCODE (Normal Mode to User Mode)	1			ms

Note11: The ReadEEPROM operation needs to operate at F_{SCLKRE} or lower clock frequency. Operation beyond this limit may cause a malfunction of reading out.

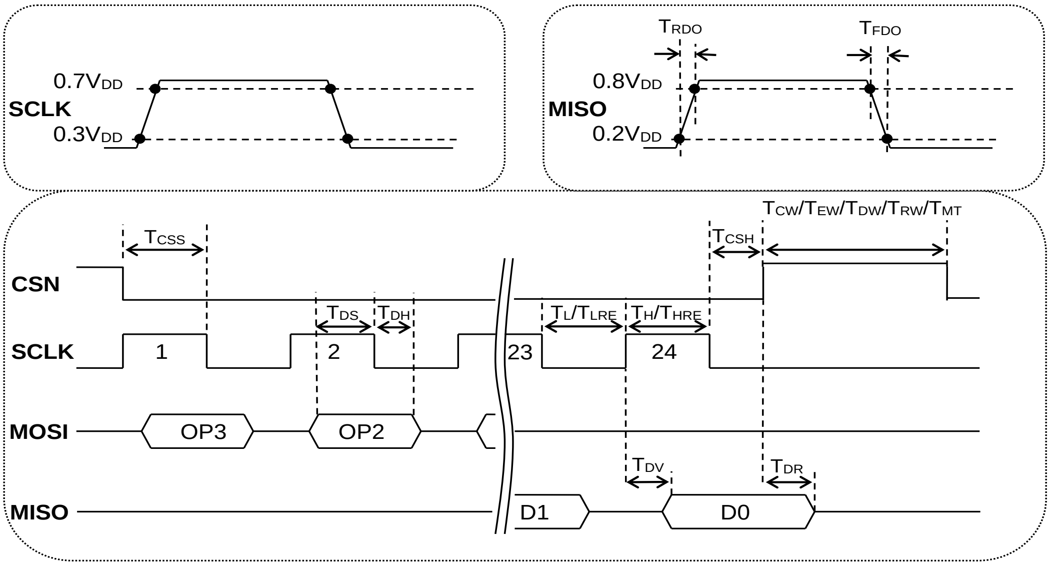


Figure3: Serial I/F AC Timing Diagram

13. ABZ Incremental Output

The AK7455 provides the ABZ incremental outputs via A, B, Z pin. These outputs are encoded from the angle output data. The output resolution is programmable from 1 to 4096ppr. When the magnetic field is rotating in counter-clockwise at default settings, the B-phase leads the A-phase.

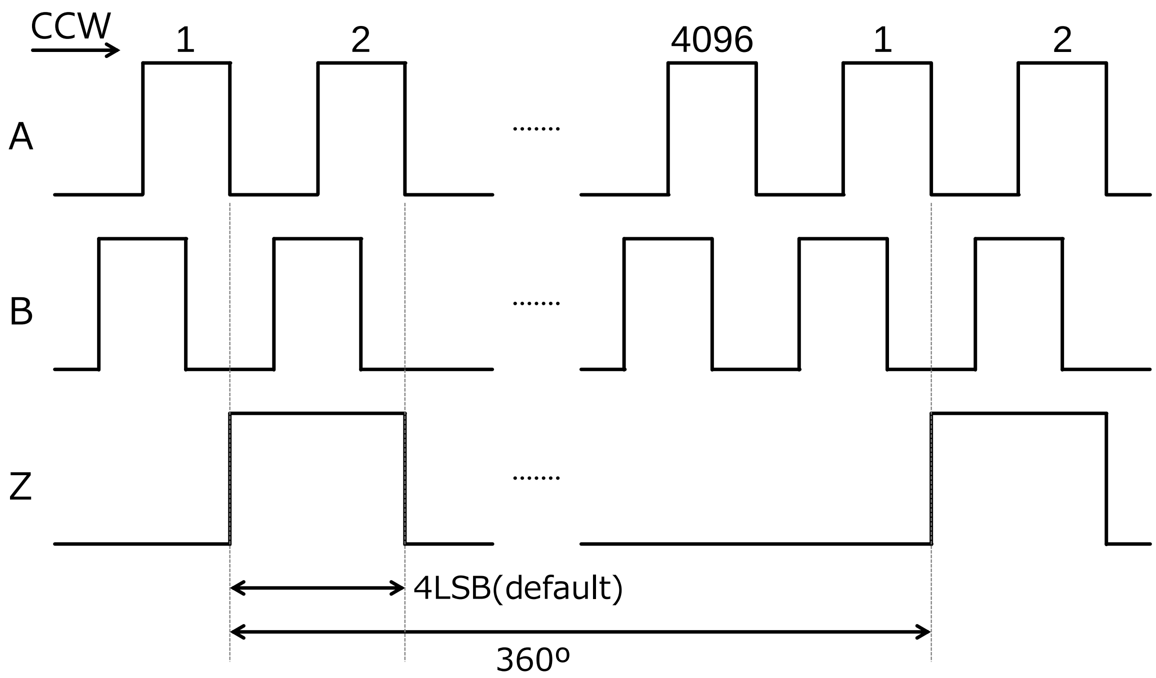


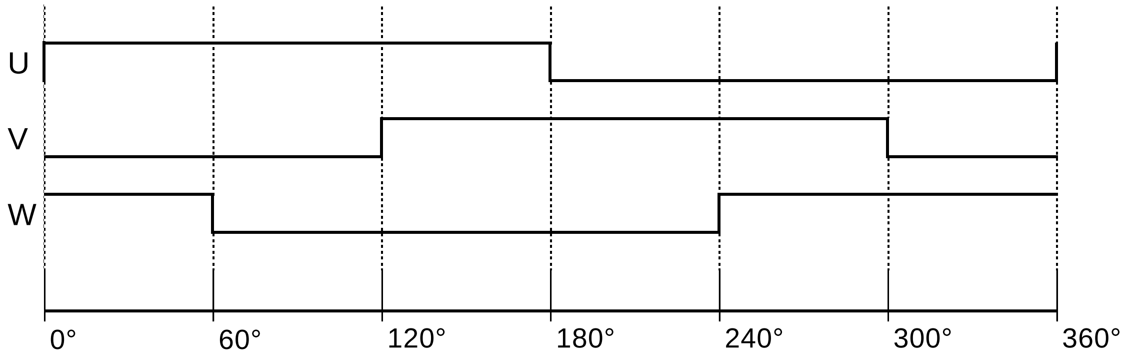
Figure4: ABZ Incremental Output

Also, the user programming function allows the Z-phase pulse width to be set to any value from 1 to 16384 times, with 1/4 of the A-phase period as the reference width.

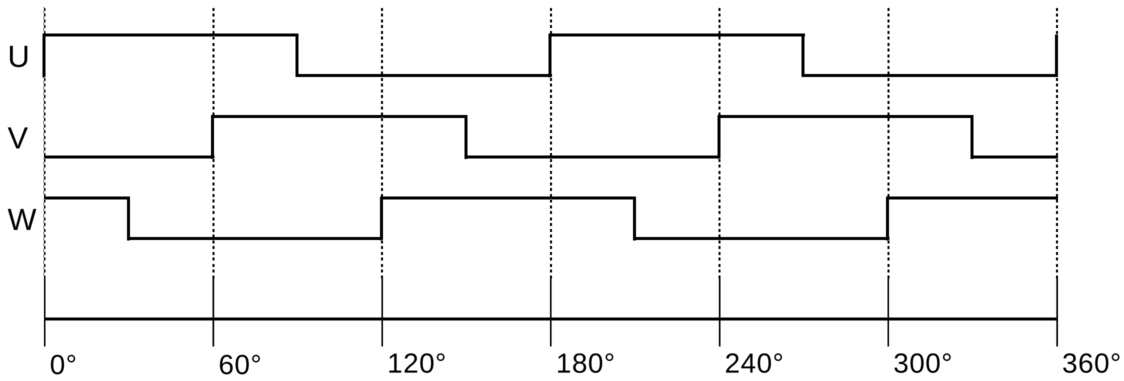
Besides, the rising edge of the Z-phase pulse can be synchronized with A-phase pulse falling, B-phase pulse rising, A-phase pulse rising, or B-phase pulse falling.
In any setting, the rising edge of the Z-phase pulse matches the absolute angle output 0º obtained by the 4-wire serial interface.

14. UVW Commutation Output

The AK7455 provides the UVW commutation output necessary for detecting the magnetic pole of the DC brushless motor via U, V, W pin. The phase difference of each output is 120º of an electric angle. The output resolution is programmable from 2 to 64poles (1 to 32pole pairs).



2 poles (1 pole pair)



4 poles (2 pole pairs)

Figure5: UVW Commutation Output

15. Programmable Mode

The AK7455 has two modes (Normal Mode and User Mode). In Normal mode, ABZ incremental output and UVW commutation output are obtained, and angle data can be read by the serial interface.

In User Mode, it can read and write the internal registers and EEPROM according to a serial interface.
User Mode supports the user programming such as zero-point adjustment, self-diagnostic ON/OFF setting, rotation direction, ABZ output setting, UVW output setting, hysteresis setting, type2 tracking servo loop filter bandwidth setting, EEPROM memory lock setting, angle INL error calibration setting, anomaly magnetic field detection setting and direction of magnetic field detection setting.
Zero-point adjustment and hysteresis setting can be set independently of ABZ incremental and UVW commutation output.

When the power supply is turned ON, the AK7455 automatically resets the internal register and load the EEPROM configuration data to set the internal configuration register. After the startup sequence, the AK7455 operates in Normal Mode.

- **Mode Description**

Each mode can be changed by writing a specific OPCODE and DATA on a specific address, as the diagram below.

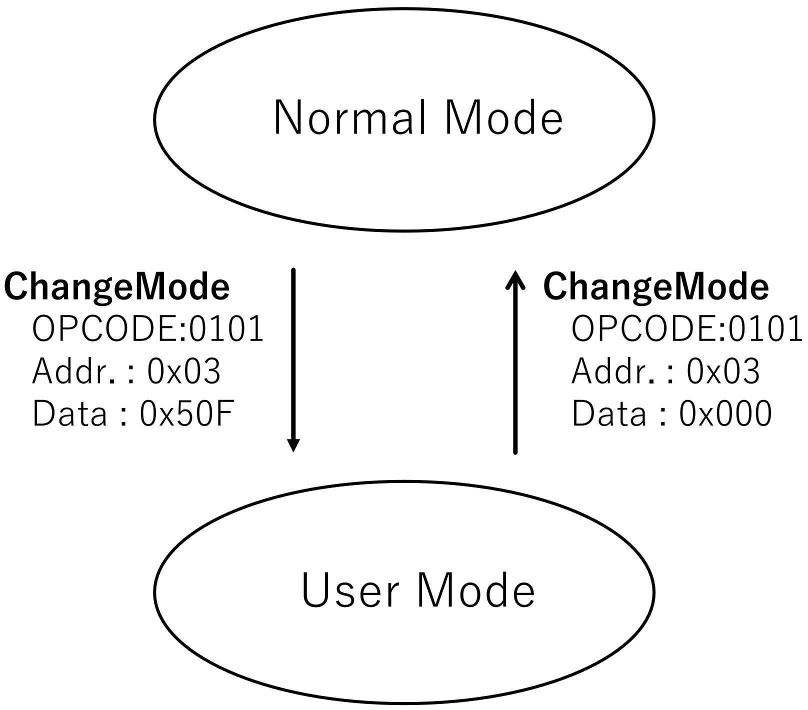


Figure6: Mode transition diagram

Mode Name	Contents
Normal Mode	The OPCODEs available are Normal Mode is only ReadAngle and ChangeMode. It is not able to access the internal register and EEPROM except register R_CHMD (Addr.0x03). The ABZ Incremental outputs and UVW commutation outputs are toggled corresponding to the magnetic field rotation. When anomaly states are detected by the self-diagnosis function, the AK7455 automatically notifies the diagnosis status by an error bit attached to the angle data output from the serial interface and Error pin output. When changing to Normal Mode, all registers are automatically initialized and loads the EEPROM configuration data to set the internal configuration register. At this time, it is necessary to secure the waiting time T_{MT} described in 12.2 AC characteristics.
User Mode	In User Mode, internal registers and EEPROM can be accessed. If you want to use the following setting in Normal Mode, it is necessary to store unique data in EEPROM in User Mode. 1. **Zero Point Setting** This setting defines the origin of the output angle data. It is programmable at any angle position in 14bit resolution. The zero-point setting can be set independently for ABZ incremental output and UVW commutation output. The absolute angle output via the serial interface is linked to the zero-point setting of the ABZ incremental output. 2. **Self-diagnostic ON/OFF Setting** When the magnetic field applied to the AK7455 reaches under 15mT (typ) or when the lock state by the type2 servo is released, or when an anomaly magnetic field is detected an error notification is sent from the ERROR pin and serial interface. Also, it is possible to set this function ON/OFF. 3. **Rotation Direction Setting** This setting defines the increase or decrease of the output angle data relative to the magnetic field rotation direction. When “CCW” is selected, the output angle data increases in response to the counter-clockwise direction of magnetic field rotation. When “CW” is selected, the output angle data increases in response to the clockwise direction of magnetic field rotation. 4. **Z-phase Pulse Phase Setting** This setting defines the phase of the Z-phase pulse in ABZ incremental output. The rising edge of the Z-phase pulse can be selected from a total of 4 rising/falling edges of the A-phase pulse/the B-phase pulse. 5. **ABZ Incremental Output ON/OFF Setting** This setting defines whether ABZ incremental output is enabled/disabled. When “OFF” is selected, the A, B, and Z pins go to high-impedance (Hi-Z) output. 6. **ABZ Incremental Output Hysteresis Setting** This setting defines the ABZ hysteresis parameter. The ABZ output is not updated when the processing data step is smaller than the programmed parameter.

	7. **Z-phase Pulse Width Setting** This setting defines the Z-phase pulse width in the ABZ incremental output. Z-phase pulse width can be set to any value from 1 to 16384 LSB within the range of the ABZ incremental output resolution setting. 8. **ABZ Incremental Output Resolution Setting** This setting defines the ABZ output resolution. It is programmable to an arbitrary value from 1 to 4096ppr. 9. **UVW Commutation Output ON/OFF Setting** This setting defines whether the UVW commutation output is enabled/disabled. When “OFF” is selected, the U, V, and W pins go to high-impedance (Hi-Z) output. 10. **UVW Commutation Output Hysteresis Setting** This setting defines the UVW output hysteresis parameter. The UVW output is not updated when the processing data step is smaller than the programmed parameter. 11. **UVW Commutation Output Resolution Setting** This setting defines the UVW output resolution. It is programmable to an arbitrary value from 2 to 64poles (1~32pole pairs). 12. **Filter Gain and Band Width Setting** The gain and band-width settings of type 2 servo tracking loop filter are possible. 13. **Memory Lock Setting** To prevent rewriting EEPROM incorrectly, it can lock the memory. Once the configuration of the memory lock is enabled, it is not possible to change EEPROM anymore, including the memory lock setting itself. 14. **Angle INL Error Calibration Setting** It is possible to calibrate the angle INL error caused by the misalignment position in the Shaft-End configuration and the angle INL error occurring when using in the Off-Axis configuration by writing the calibration parameter in the specific memory. 15. **Anomaly Magnetic Field Detection Function Setting** It is possible to set up the anomaly magnetic field detection function and to reduce the angular error caused by anomaly magnetic fields. 16. **Magnetic field direction to be detected Setting** The direction of the magnetic field (XY, XZ, and YZ axes) can be set. In User Mode, the ERROR pin goes to the “L” level output. In User Mode, the magnetic flux density data and self-diagnostic data are not updated automatically. To obtain the latest data, send OPCODE: DataRenew in advance, and read each data after a wait time of 150µs(min). At the time of transition to User Mode, it is necessary to secure the waiting time T_{MT} described in 12.2 AC characteristics. **WARNING:** Electrical and magnetic characteristics are not guaranteed in User Mode.

16. Serial I/F Protocol

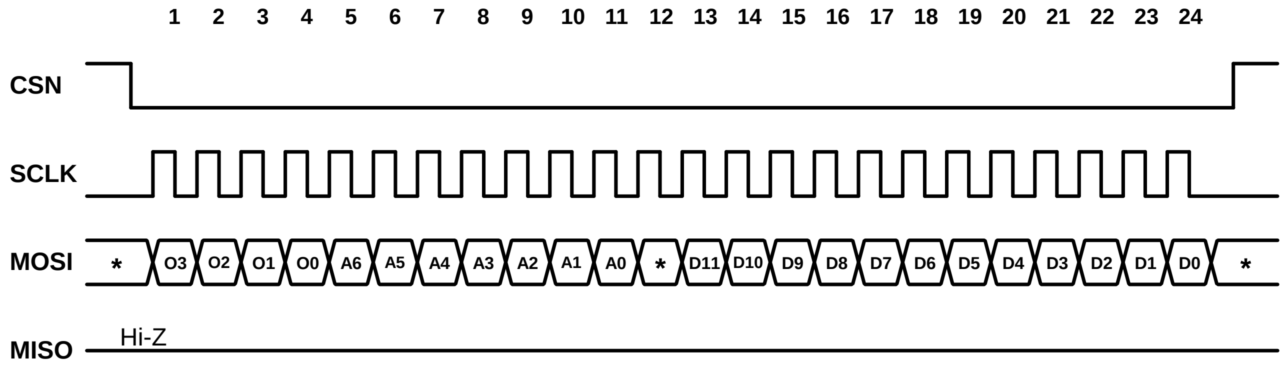
The AK7455 provides the 4-wire serial interface for user programming. Data communication is available only when the CSN pin sets to “L”. To write the internal register or EEPROM, serial data must be input via the MOSI pin on the falling edge of SCLK. To read the internal register or EEPROM, the AK7455 outputs serial data via the MISO pin on the rising edge of SCLK.

➢ **OPCODE**

OPCODE [3:0]	Code name	Description	Permission	
			Normal Mode	User Mode
0000	N/A	N/A	invalid	invalid
0001	WriteEEPROM	Write data to EEPROM	invalid	**valid**
0010	ReadEEPROM	Read data from EEPROM	invalid	**valid**
0011	WriteReg	Write data to the internal register	invalid	**valid**
0100	ReadReg	Read the internal register data	invalid	**valid**
0101	ChangeMode	Change operating mode	**valid**	**valid**
0110	N/A	N/A	invalid	invalid
0111	N/A	N/A	invalid	invalid
1000	DataRenew	Update the processing data	invalid	**valid**
1001	ReadAngle	Read the angle data	**valid**	**valid**
1010	N/A	N/A	invalid	invalid
1011	N/A	N/A	invalid	invalid
1100	N/A	N/A	invalid	invalid
1101	N/A	N/A	invalid	invalid
1110	N/A	N/A	invalid	invalid
1111	N/A	N/A	invalid	invalid

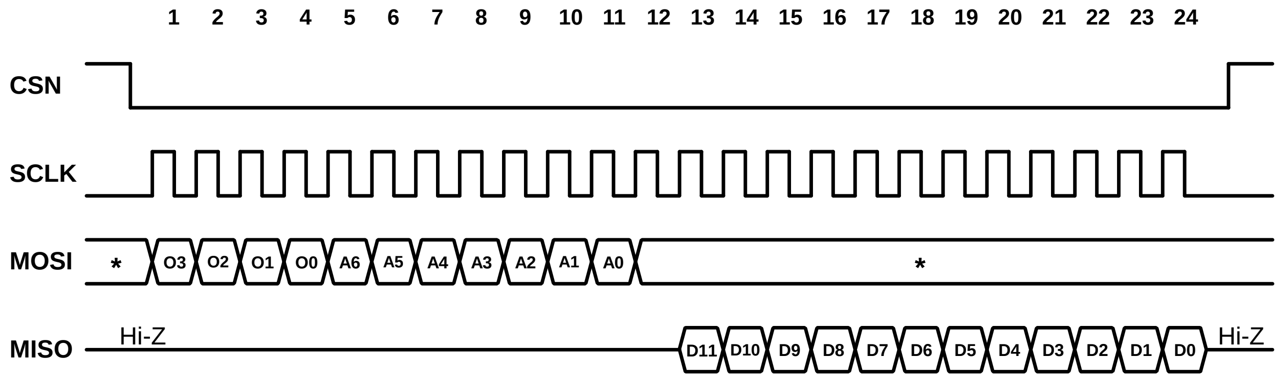
➢ Data Format

1) Write data (WriteEEPROM or WriteReg)



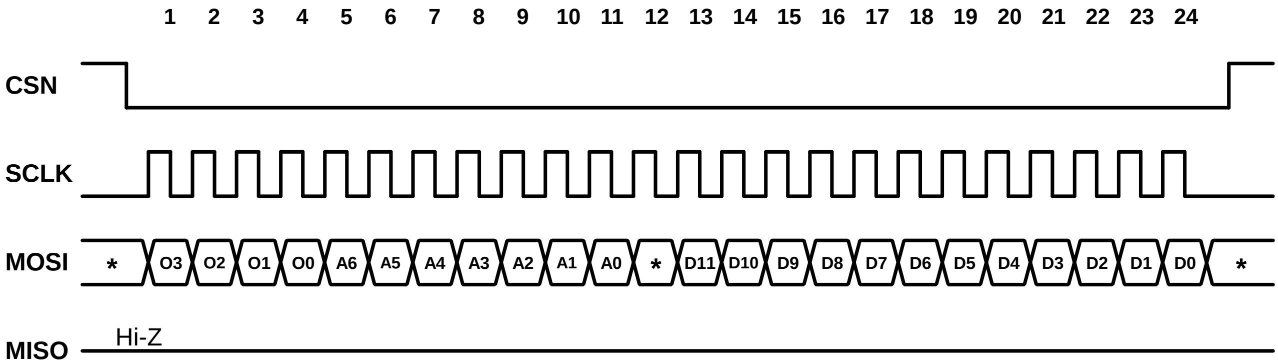
OPCODE(O[3:0])=0001(WriteEEPROM)
OPCODE(O[3:0])=0011(WriteReg)
A[6:0]:Register or EEPROM address
D[11:0]:Writing data

2) Read Data (ReadEEPROM or ReadReg)



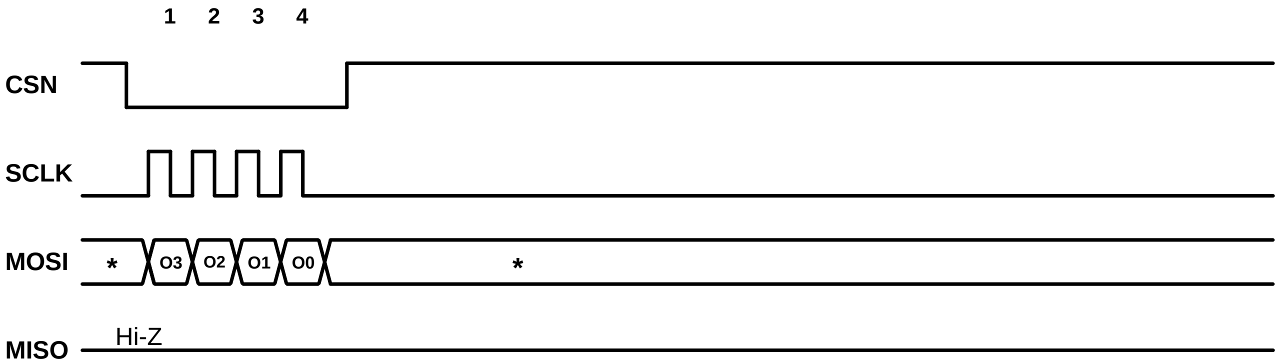
OPCODE(O[3:0])=0010(ReadEEPROM)
OPCODE(O[3:0])=0100(ReadReg)
A[6:0]:Register or EEPROM address
D[11:0]:Reading data

3) ChangeMode



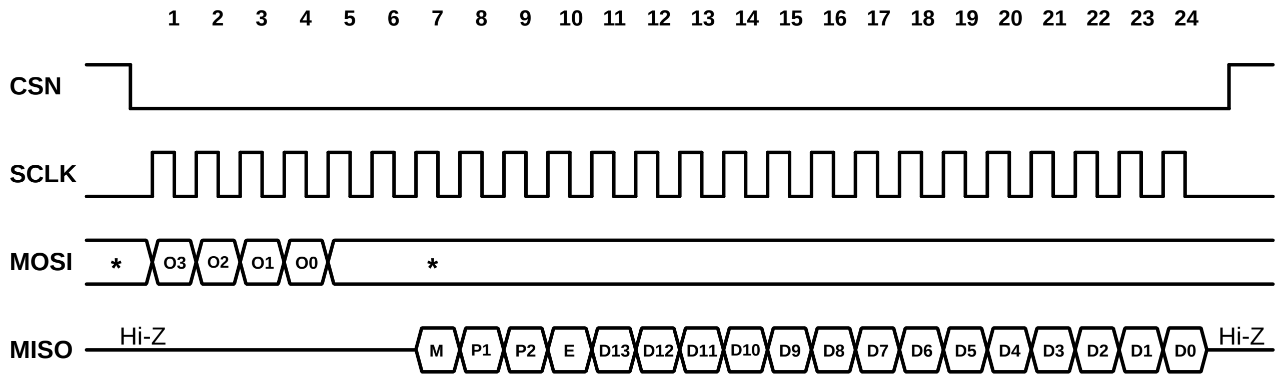
OPCODE(O[3:0])=0101
A[6:0]:0x03
D[11:0]:“0x000”(Normal Mode), “0x50F”(User Mode)

4) DataRenew



OPCODE(O[3:0])=1000

5) ReadAngle



OPCODE(O[3:0])=1001
M: Mode bit (Normal Mode="0", User Mode="1")
P1: Parity bit for angle data D[13:7] (Normal Condition: ODD Parity, Anomaly Condition: EVEN Parity)
P2: Parity bit for angle data D[6:0] (Normal Condition: ODD Parity, Anomaly Condition: EVEN Parity)
E: Error bit (Normal Condition="1", Anomaly Condition="0")
D[13:0]:14bit angle data
The correspondence between the angle data value and angle position is as follows.

Angle Position[°]	**D[13:0]**
0	0x0000
0.022	0x0001
0.044	0x0002
0.066	0x0003
-	-
89.978	0x0FFF
90.000	0x1000
-	-
359.956	0x3FFE
359.978	0x3FFF

17. Self-diagnostic Function

The AK7455 has the following self-diagnostic functions.

- **Low Magnetic Flux Density Detection**
  The AK7455 indicates an error if the applied magnetic flux density is less than 15mT(typ).
  This error sets the ERROR pin to “L” level and error bit to “0”.
  The diagnosis result is updated every 62.5μs(typ). Once the low magnetic field state is resolved after the error signal is issued, the error signal will be canceled after 0.25ms(typ).

  WARNING: In the case of Off-Axis configuration, it is assumed that the magnetic flux density will be below 15mT even under normal conditions, so it is recommended to disable this function.

- **Loss of Tracking Detection**
  The AK7455 indicates an error when there is a loss of tracking over 2º(typ). This error sets the ERROR pin to “L” level and error bit to “0”.
  The diagnosis result is updated every 62.5μs(typ). Once the tracking lost state is resolved after the error signal is issued, the error signal is canceled after 12ms(typ).

- **Anomaly magnetic field detection**
  A magnetic field other than the sensor magnet applied to the AK7455 will be judged as an anomaly, and an error will be issued. This error sets the ERROR pin to "L" level and the error bit to "0".
  The magnetic field state is sampled every 45° of rotation, and the angular error information is updated and reported with one rotation of the magnet.
  If no anomaly is detected during one rotation of the magnet, the error reporting is canceled at that point.

If neither diagnostic result is an error, the AK7455 automatically sets the ERROR pin to the “H” level and sets the error bit to “1”.

18. Anomaly magnetic field detection

The AK7455 has a function of detecting an anomaly magnetic field other than that of the sensor magnet and issuing an ERROR pin and register.
If a magnetic field other than the sensor magnet inputs the direction of the AK7455's magnetic field, it will cause an angular error.
This function defines the magnetic field other than the sensor magnet as an anomaly magnetic field.
If the angular error estimated from the anomaly magnetic field exceeds a certain amount, the ERROR pin and register will report the anomaly condition to the user.

This function is realized by storing the magnetic field state in the absence of an anomaly magnetic field and calculating the angular error caused by the magnetic field input different from this state. When the angular error obtained by the calculation exceeds a certain threshold level, the system judges that it is an anomaly state and issues an alarm.
The angular error level at which an anomaly condition is reported can be changed by user settings.

Therefore, when using this feature, the ideal magnetic field condition must be stored in the AK7455 as the default setting (Note12).

It also has a function to dynamically calibrate the angular error caused by anomalous magnetic fields. The angular error can be reduced dynamically by updating the calibration parameters to reduce the angular error with each revolution of the magnet (Note12).

Note12: This datasheet does not provide information on how to set up this function. Please contact us for more information.

19. Register Address Map / Description

➢ Register Address Map

Address	Register Name	Permission		Contents
		Normal Mode	User Mode	
0x01	R_DVID[11:0]	N/A	R	AK7455 device ID
0x02	R_MAG[11:0]	N/A	R	Magnetic flux density data
0x03	R_CHMD[11:0]	W	R/W	Mode indicator data
0x04	R_ERRMON[5:0]	N/A	R	Diagnosis error monitor
0x05	R_ZP_U[1:0]	N/A	R/W	14bit Zero-point setting for ABZ and SPI (Upper 2bit)
0x06	R_ZP_L[11:0]	N/A	R/W	14bit Zero-point setting for ABZ and SPI (Lower 12bit)
0x07	R_RDABZ[11:0]	N/A	R/W	Diagnostic function setting Rotation direction setting Z-phase pulse phase setting ABZ output ON/OFF setting ABZ output hysteresis setting
0x08	R_ZW_U[1:0]	N/A	R/W	Z-phase pulse width setting (Upper 2bit)
0x09	R_ZW_L[11:0]	N/A	R/W	Z-phase pulse width setting (Lower 12bit)
0x0A	R_ABZ_RES[11:0]	N/A	R/W	ABZ Output resolution setting
0x0B	R_UVW[11:0]	N/A	R/W	UVW output ON/OFF setting UVW output hysteresis setting UVW output resolution setting 14bit Zero-point setting for UVW (Upper 2bit)
0x0C	R_ZP_UVW_L[11:0]	N/A	R/W	14bit Zero-point setting for UVW (Lower 12bit)
0x0D	R_SRV_SET[11:0]	N/A	R/W	Servo filter setting Angle INL error calibration data
0x0E	R_MLK[1:0]	N/A	R	Memory lock indicator
0x0F	R_PCAL1A[10:0]	N/A	R/W	Angle INL Error calibration data
0x10	R_PCAL1P[11:0]	N/A	R/W	
0x11	R_PCAL2A[10:0]	N/A	R/W	
0x12	R_PCAL2P[11:0]	N/A	R/W	
0x13	R_PCAL3A[10:0]	N/A	R/W	
0x14	R_PCAL3P[11:0]	N/A	R/W	
0x15	R_PCAL4A[10:0]	N/A	R/W	
0x16	R_PCAL4P[11:0]	N/A	R/W	
0x17	R_CA[11:0]	N/A	R/W	
0x18	R_GM[11:0]	N/A	R/W	
0x19	R_FSMF[11:0]	N/A	R/W	The direction of magnetic field detection, anomaly magnetic field detection, and error reduction settings
0x1A	R_SIN1_INIT[8:0]	N/A	R/W	Anomaly magnetic field detection and error reduction settings
0x1B	R_COS1_INIT[8:0]	N/A	R/W	
0x1C	R_SIN2_INIT[8:0]	N/A	R/W	
0x1D	R_COS2_INIT[8:0]	N/A	R/W	
0x1E	R_SIN1[8:0]	N/A	R	Reading data to reduce errors due to anomaly magnetic fields
0x1F	R_COS1[8:0]	N/A	R	
0x20	R_SIN2[8:0]	N/A	R	

0x21	R_COS2[8:0]	N/A	R	
0x22	R_CALRDY	N/A	R	READY flag for reading initial data of anomaly magnetic field detection

(N/A: not available, R: read-only, W: write-only, R/W: full access)

➢ Register description

1) R_DVID (Addr. 0x01)

(Addr.0x01) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_DVID[11:0]											

The R_DVID register is the device ID. The AK7455 is 0x155.

2) R_MAG (Addr. 0x02)

(Addr.0x02) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_MAG[11:0]											

The R_MAG register contains the 12bit magnetic flux density data. The data resolution is 30μT/LSB(typ).
In User Mode, the magnetic flux density can be detected by reading these registers.
DataRenew command needs to be sent in advance to get the latest data.
When manufacturing a rotary angle sensor module using magnets, it is possible to check whether the appropriate magnetic flux density is applied. The relationship between the data value and angle position is as follows.
The magnetic flux density measured by this function is the square root of the sum of the squares of the magnetic flux density of the measurement axes (2 axes) set by the R_FIELDSEL register.

Magnetic Flux Density[mT]	R_MAG[11:0]
0	0x000
0.03	0x001
0.06	0x002
0.09	0x003
-	-
122.85	0xFFF

WARNING: This magnetic flux density measurement data's accuracy which is stored in this register is not guaranteed. It is recommended to use only for reference.

3) R_CHMD (Addr. 0x03)

(Addr.0x03) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_CHMD[11:0]											

The R_CHMD register contains the 12bit mode indication data. In Normal mode, mode change can be done by using ChangeMode OPCODE and storing instruction data in the R_CHMD register.

Mode	R_CHMD[11:0]	Default
Normal Mode	0x000	●
User Mode	0x50F	

4) R_ERRMON (Addr. 0x04)

(Addr.0x04) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-						R_ERRMON[5:0]					

The R_ERRMON register contains the diagnostic indication data. Each indication data bit is set to “0” when an anomaly state is detected. The relationship between each item and the bits is as follows.

R_ERRMON [5:0]	Diagnosis	Anomaly Condition	Normal Condition
R_ERRMON[5]	Angular error estimated from the detection field is greater than 5.6º(typ) (Note13)	0	1
R_ERRMON[4]	The angular error estimated by subtracting the initial magnetic field from the detection field is greater than the R_CAL_RANGE setting value.		
R_ERRMON[3]	Reaching the upper and lower limits of calibration parameters for error reduction by anomalous magnetic fields.		
R_ERRMON[2]	*AKM reserved*		
R_ERRMON[1]	Low Magnetic Flux Density Detection		
R_ERRMON[0]	Tracking Lost State Detection		

The magnetic field is sampled and updated every revolution of the magnet.
R_ERRMON[5:3] updates the judgment of every revolution of the magnet.
When R_MAGCAL=0, each bit of R_ERRMON[5:3] is always “1”.
When R_BACKCAL=0, R_ERRMON[3] is always “1”.

Note13: This value is for Shaft-End configuration. For Off-Axis configuration, it depends on the accuracy of angle INL error calibration. Please contact us for more information.

5) R_ZP (Addr. 0x05, 0x06)

(Addr.0x05) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										R_ZP_U[1:0]	

(Addr.0x06) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_ZP_L[11:0]											

The R_ZP register contains the 14bit configuration data of zero-degree point. This register is used to set an arbitrary angle position to zero (angle output 0º).

Zero Point Position[º]	R_ZP_U[1:0]	R_ZP_L[11:0]	Default
0	0x0	0x000	●
0.022	0x0	0x001	
0.044	0x0	0x002	
0.066	0x0	0x003	
-	-	-	
89.978	0x0	0xFFF	
90.000	0x1	0x000	
-	-	-	

359.956	0x3	0xFFE	
359.978	0x3	0xFFF	

Note14: The value set in this register is reflected in SPI absolute angle output and ABZ output. The zero-degree point position for the UVW phase is set by R_ZP_UVW.

6) R_RDABZ (Addr. 0x07)

(Addr.0x07) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_DIS_ERR [3:0]				R_RD	R_Z_PHASE [1:0]		R_ABZ_E	R_ABZ_HYS[3:0]			

The R_RDABZ register contains enable/disable setting data of diagnosis function, rotation direction setting data, Z-phase pulse phase setting data, ABZ phase output ON/OFF setting data, and ABZ phase output hysteresis setting data.

6-1. R_DIS_ERR

The R_DIS_ERR register stores ON/OFF setting data of each self-diagnosis function. To turn off the self-diagnosis function, set the corresponding bit to “1”.

R_DIS_ERR [3:0]	**Diagnosis**	**Default**
R_DIS_ERR[3]	Anomaly magnetic field detection Applies to the detection of R_ERRMON[5:3]	0
R_DIS_ERR[2]	*AKM reserved*	0
R_DIS_ERR[1]	Low magnetic flux density detection Applies to the detection of R_ERRMON[1]	0
R_DIS_ERR[0]	Tracking lost state detection Applies to the detection of R_ERRMON[0]	0

Note15: Please set R_DIS_ERR[2] to 0. To enable the setting of R_DIS_ERR[3], in addition to the above, it is necessary to set R_MAGCAL=1.

6-2. R_RD

The R_RD register stores the configuration data of the rotation direction. It defines the increase or decrease of the output angle data relative to the magnetic field rotation direction.

Even if the rotation position of the magnet is the same in the Shaft-End and the Off-Axis configuration, the angle output by the AK7455 differs because the direction of the magnetic field applied to the AK7455 is different.

[Shaft-End Configuration]

When selecting CCW (counter-clockwise), the output angle data increases in response to counter-clockwise magnet rotation. Counter-clockwise is defined by the 1-6-7-12-13-18-19-24 pin order direction in a top view of the QFN-24 package.

When selecting CW (clockwise), the output angle data increases in response to clockwise magnet rotation. Clockwise is defined by the 24-19-18-13-12-7-6-1 pin order direction in a top view of the QFN-24 package.

Rotation Direction	**R_RD**	**Default**
CCW	0x0	●
CW	0x1	

[Off-Axis Configuration]
In the case of R_RD=0 (CCW), R_ZP_U[1:0]=0x0, R_ZP_L[11:0]=0x0, the direction of the magnetic field and the angle output value for each arrangement of the AK7455 at each magnet rotation angle are shown in Figure7.
Figure 8 shows the relationship between the angle of rotation and the angle output of the AK7455 in each of the configurations shown in Figure 7.

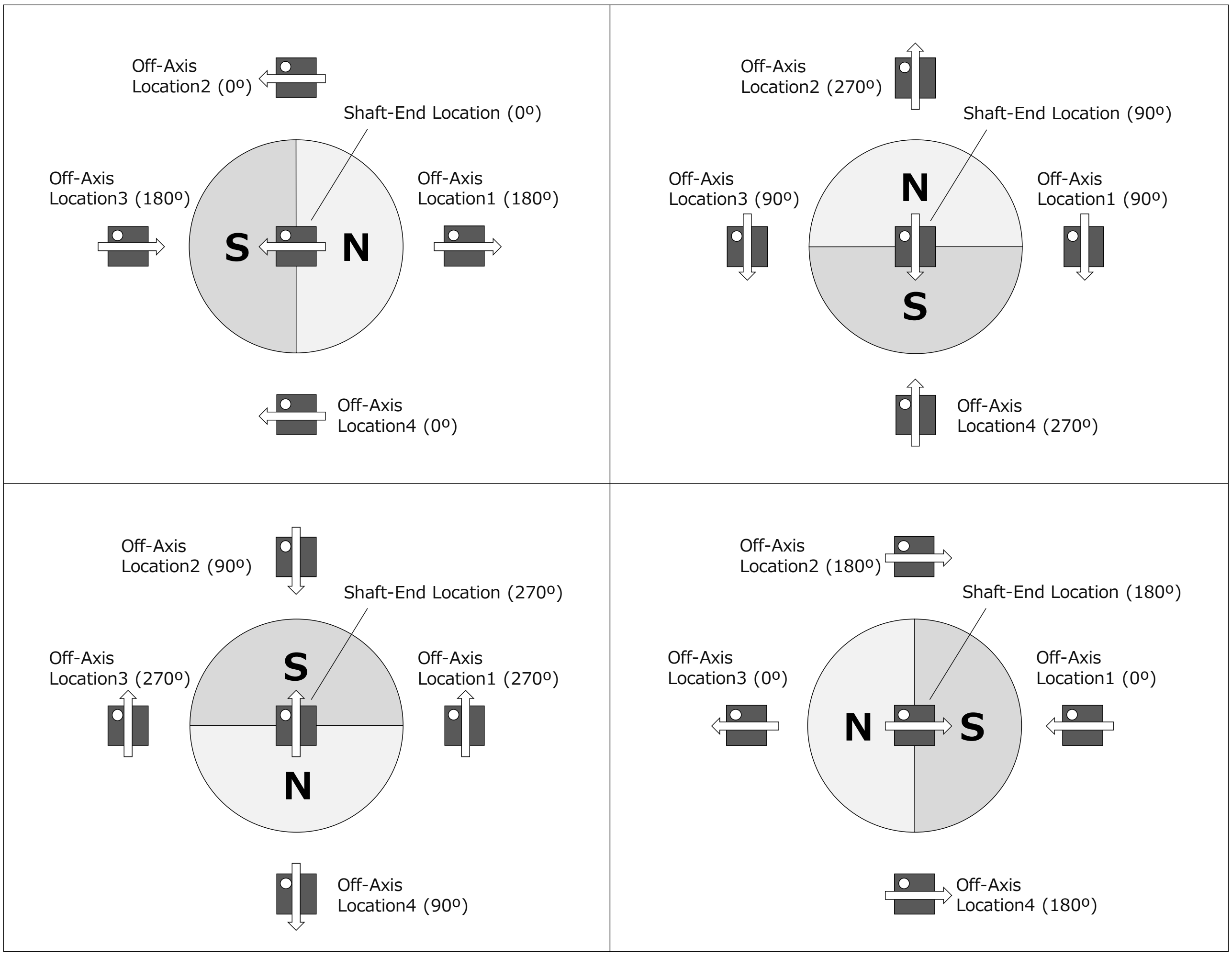


Figure7: Relationship between magnet position and angle in Shaft-End and Off-Axis

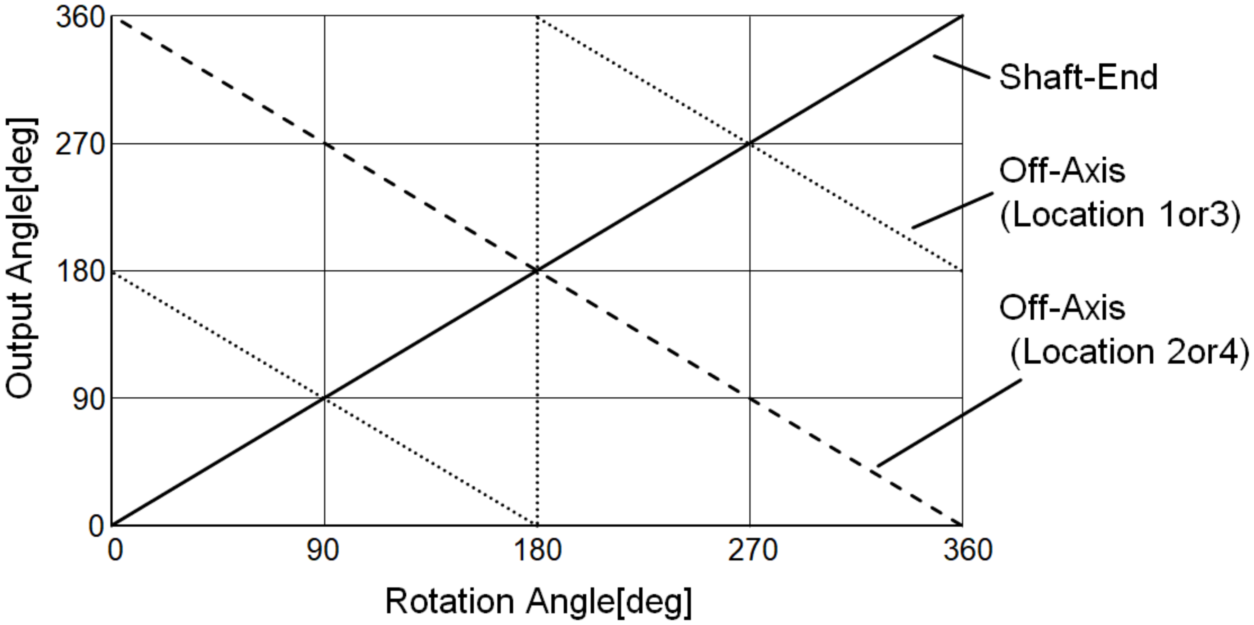


Figure8: Relationship between magnet rotation angle and output angle in different arrangements

6-3. R_Z_PHASE

The R_Z_PHASE register is the Z-phase pulse phase setting data. The rising edge of the Z-phase pulse can be synchronized with the following four types of edges.
Regardless of the setting, the Z-phase pulse rising edge is generated at an absolute angle of 0º obtained by SPI communication.

Z-phase Pulse Rising Edge	R_Z_PHASE[1:0]	Default
A-phase Pulse Falling Edge	0x0	●
B-phase Pulse Rising Edge	0x1	
A-phase Pulse Rising Edge	0x2	
B-phase Pulse Falling Edge	0x3	

The Figure9 shows the relationship between the R_Z_PHASE setting value and the Z-phase pulse when R_RD=0, R_ZW_U=0, R_ZW_L=0, and the magnetic field rotation direction is CCW.

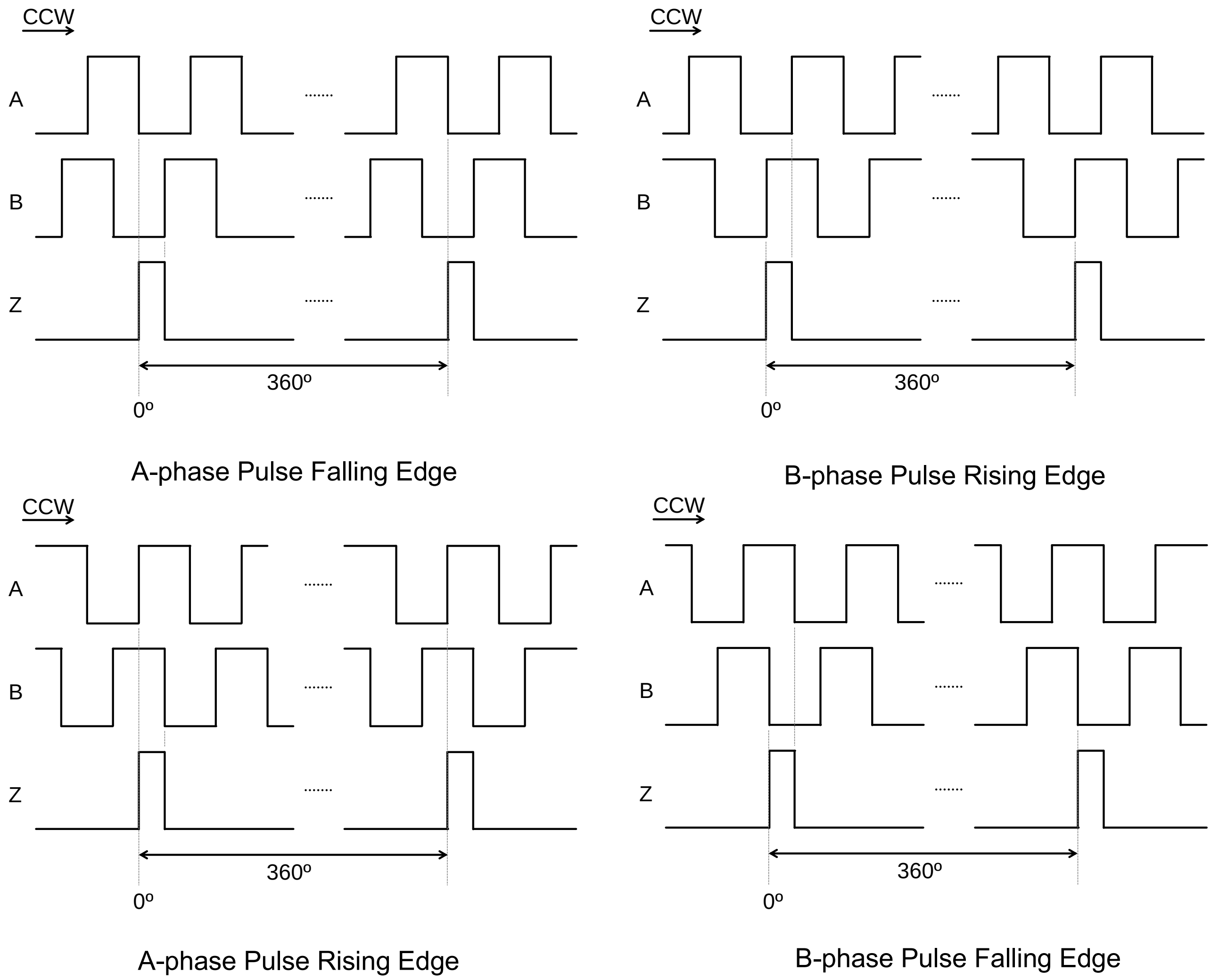


Figure9: Z-Phase Pulse Phase Setting

6-4. R_ABZ_E
The R_ABZ_E register stores the configuration data of ABZ incremental output ON/OFF.
When selecting “OFF”, A, B, and Z pins go to high-impedance (Hi-Z) output.

WARNING: During the start-up time after power-on and the transition from User Mode to Normal Mode (max: 40ms), the A, B, and Z pins will be “L” regardless of this setting.

ABZ Output	R_ABZ_E	Default
OFF	0x0	
ON	0x1	●

6-5. R_ABZ_HYS

The R_ABZ_HYS register stores the configuration data of the hysteresis filter parameter.
This parameter defines whether the output angle data is updated or not in response to the processing of the raw angle data change. The output angle data is not updated when the processing of the raw angle data change is smaller than the hysteresis width. This parameter is defined in units of 1LSB of 14bit angle data.
When the hysteresis width is increased, the data oscillation influence caused by a noisy environment (ex. module vibration) is decreased. However, the maximum output delay is increased as described below equation.

$$\text{Maximum Output Delay } T_D = 1.2 + 0.125(1+\text{"ABZ Hysteresis"})\ [\mu s]$$

Note16: “ABZ Hysteresis” in the above equation is the number of LSB set by R_ABZ_HYS[3:0]. (Example: “ABZ Hysteresis" = 3 when R_ABZ_HYS[3:0]=0x4.)

When selecting the “OFF” setting, the output angle data equals the processing of the raw angle data. The inherent LSB data flickering (Note17) is not eliminated from the output angle data. The maximum output delay is the shortest (1.2μs) in this setting.
When selecting the “0LSB” setting, the inherent LSB data flickering is eliminated from the output angle data. The maximum output delay is 1.325μs in this setting.

Note17: The LSB data flickering inheres in the type2 tracking loop architecture. This data flickering appears even in the ideal environment under a noise-free case. Therefore, AKM does not recommend the “OFF” setting.

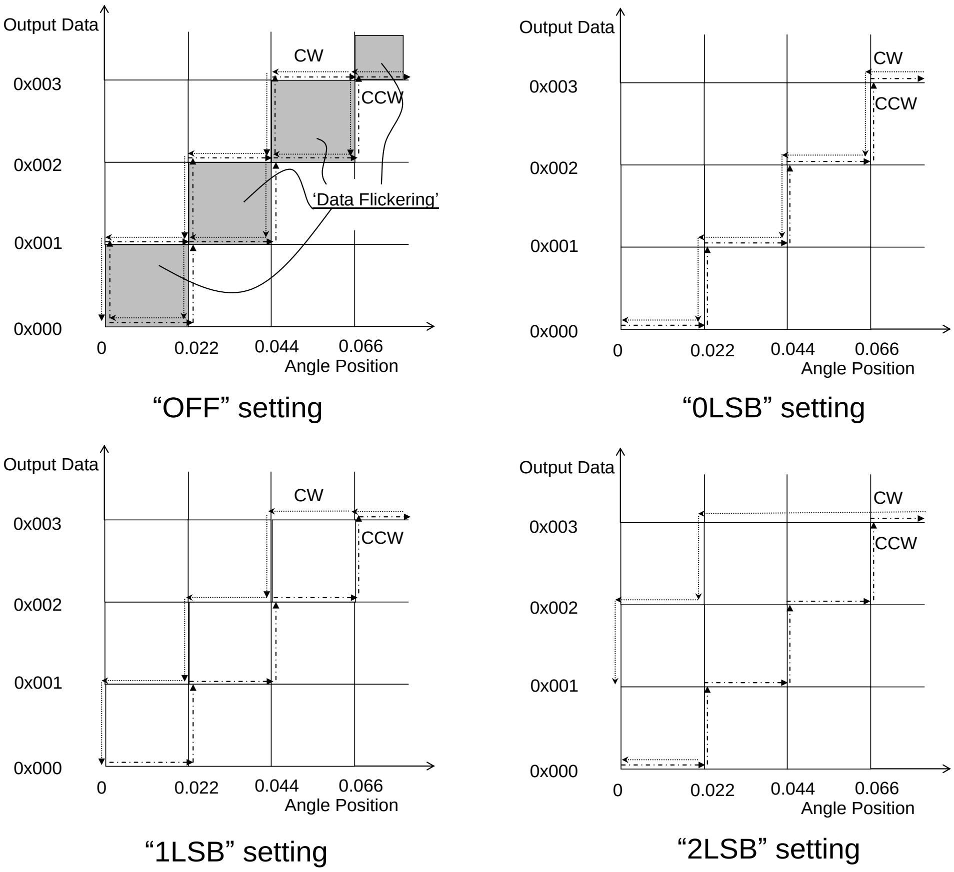


Figure10: ABZ Hysteresis Setting

The relationship between R_ABZ_HYS[3:0] and the hysteresis set value is as follows.

ABZ Hysteresis	R_ABZ_HYS[3:0]	Default
OFF	0x0	
0LSB	0x1	
1LSB	0x2	●
2LSB	0x3	
3LSB	0x4	
4LSB	0x5	
5LSB	0x6	
6LSB	0x7	
7LSB	0x8	
8LSB	0x9	
9LSB	0xA	
10LSB	0xB	
11LSB	0xC	
12LSB	0xD	
OFF	0xE	
OFF	0xF	

7) R_ZW (Addr. 0x08, 0x09)

(Addr.0x08) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										R_ZW_U[1:0]	

(Addr.0x09) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_ZW_L[11:0]											

The R_ZW register contains the Z-phase pulse width setting data of the ABZ incremental output.
It can be set to any value from 1 to 16384LSB. 1LSB is 1/4 of A phase one cycle set by R_ABZ_RES (ABZ phase resolution setting).
When the setting value satisfies the condition of R_ZW+1≧4(R_ABZ_RES+1), the Z-phase pulse always goes to the "H" level.

Z-phase Pulse Width[LSB]	R_ZW_U[1:0]	R_ZW_L[11:0]	Default
1	0x0	0x000	
2	0x0	0x001	
3	0x0	0x002	
4	0x0	0x003	●
-	-	-	
4096	0x0	0xFFF	
4097	0x1	0x000	
-	-	-	
16383	0x3	0xFFE	
16384	0x3	0xFFF	

Figure11 shows an example of the Z-phase pulse width when R_RD=0x0, R_ABZ_RES=0x9C3 (Default), and the magnetic field rotation direction is CCW.

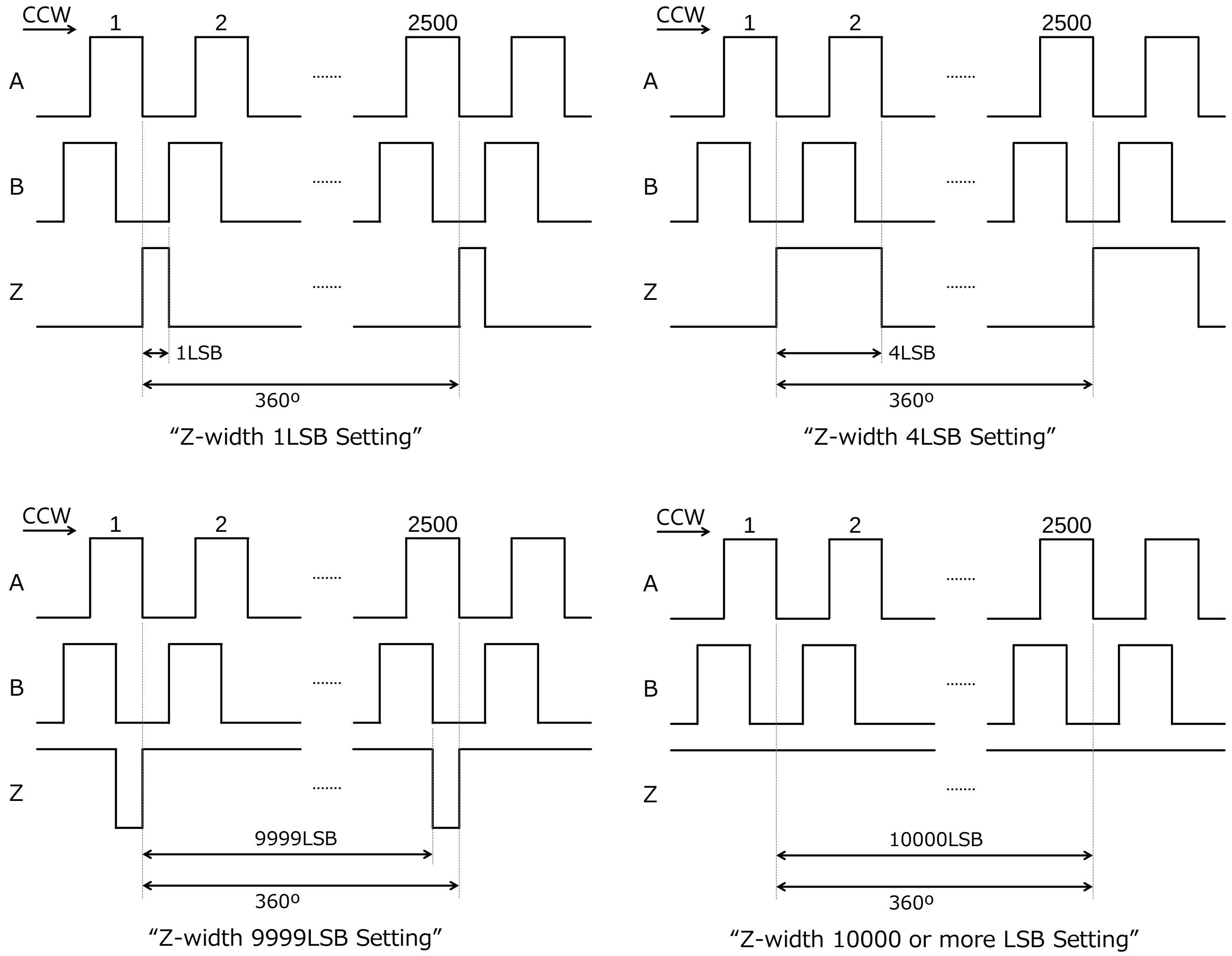


Figure11: Z-phase Pulse Width Setting

8) R_ABZ_RES (Addr. 0x0A)

(Addr.0x0A) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_ABZ_RES[11:0]											

The R_ABZ_RES register contains the configuration data of ABZ incremental output resolution. It can be set to an arbitrary value from 1 to 4096ppr.

ABZ Resolution	R_ABZ_RES[11:0]	Default
1ppr	0x000	
2ppr	0x001	
-	-	
1000ppr	0x3E7	
-	-	
2500ppr	0x9C3	●
-	-	
4095ppr	0xFFE	
4096ppr	0xFFF	

9) R_UVW (Addr. 0x0B)

(Addr.0x0B) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_UVW_E	R_UVW_HYS[3:0]				R_UVW_RES[4:0]					R_ZP_UVW_U[1:0]	

The R_UVW register contains the configuration data of UVW output ON/OFF, UVW output hysteresis, UVW output resolution and the UVW phase 14-bit zero-point data (upper 2 bits).

9-1. R_UVW_E

The R_UVW_E register stores the configuration data of the UVW commutation output ON/OFF.
When selecting the “OFF” setting, U, V, and W pins go to high-impedance (Hi-Z) output.

WARNING: During the start-up time after power-on and the transition from User Mode to Normal Mode (max: 40ms), the U, V, and W pins will be "L" regardless of this setting.

UVW Output	R_UVW_E	Default
OFF	0x0	
ON	0x1	●

9-2. R_UVW_HYS

The R_UVW_HYS register stores the configuration data of the hysteresis filter parameter.
This parameter defines whether the UVW commutation output is updated or not in response to the processing raw angle data change. The UVW commutation output is not updated when the processing of the raw angle data change is smaller than the hysteresis parameter. This parameter is defined in units of 1LSB of 14bit angle data.

WARNING: This hysteresis parameter works independently of the R_ABZ_HYS setting.

The relationship between R_UVW_HYS[3:0] and the hysteresis setting value is as follows.

UVW Hysteresis	R_UVW_HYS[3:0]	Default
OFF	0x0	
0LSB	0x1	
1LSB	0x2	●
2LSB	0x3	
3LSB	0x4	
4LSB	0x5	
5LSB	0x6	
6LSB	0x7	
7LSB	0x8	
8LSB	0x9	
9LSB	0xA	
10LSB	0xB	
11LSB	0xC	
12LSB	0xD	
OFF	0xE	
OFF	0xF	

9-3. R_UVW_RES

The R_UVW_RES register stores the configuration data of the UVW commutation output resolution. It can be set to an arbitrary value from 2 to 64poles (1 to 32pole pairs).

UVW Resolution	R_UVW_RES[4:0]	Default
2 poles (1 pole pair)	0x00	
4 poles (2 pole pairs)	0x01	
6 poles (3 pole pairs)	0x02	
8 poles (4 pole pairs)	0x03	●
10 poles (5 pole pairs)	0x04	
12 poles (6 pole pairs)	0x05	
14 poles (7 pole pairs)	0x06	
16 poles (8 pole pairs)	0x07	
18 poles (9 pole pairs)	0x08	
20 poles (10 pole pairs)	0x09	
22 poles (11 pole pairs)	0x0A	
24 poles (12 pole pairs)	0x0B	
26 poles (13 pole pairs)	0x0C	
28 poles (14 pole pairs)	0x0D	
30 poles (15 pole pairs)	0x0E	
32 poles (16 pole pairs)	0x0F	

34 poles (17 pole pairs)	0x10	
36 poles (18 pole pairs)	0x11	
38 poles (19 pole pairs)	0x12	
40 poles (20 pole pairs)	0x13	
42 poles (21 pole pairs)	0x14	
44 poles (22 pole pairs)	0x15	
46 poles (23 pole pairs)	0x16	
48 poles (24 pole pairs)	0x17	
50 poles (25 pole pairs)	0x18	
52 poles (26 pole pairs)	0x19	
54 poles (27 pole pairs)	0x1A	
56 poles (28 pole pairs)	0x1B	
58 poles (29 pole pairs)	0x1C	
60 poles (30 pole pairs)	0x1D	
62 poles (31 pole pairs)	0x1E	
64 poles (32 pole pairs)	0x1F	

10) R_ZP_UVW_L (Addr. 0x0C)

(Addr.0x0C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_ZP_UVW_L[11:0]											

The lower 2 bits of the Addr. 0x0B: R_ZP_UVW_U[1:0] and this register contain 14-bit zero point information data.
These are registers to set the arbitrary angle position to UVW phase zero point (angle output 0º). It operates independently of the zero-point setting (R_ZP) for the ABZ phase.

Zero Point Position[º]	**R_ZP_UVW_U [1:0]**	**R_ZP_UVW_L [11:0]**	**Default**
0	0x0	0x000	●
0.022	0x0	0x001	
0.044	0x0	0x002	
0.066	0x0	0x003	
-	-	-	
89.978	0x0	0xFFF	
90.000	0x1	0x000	
-	-	-	
359.956	0x3	0xFFE	
359.978	0x3	0xFFF	

Note18: The above zero position is set with respect to the absolute angle position when R_ZP=0x0 is set.

11) R_SRV_SET (Addr. 0x0D)

(Addr.0x0D) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_SRVGAIN[7:0]								R_SRV	R_FILTER[2:0]		

11-1. R_SRVGAIN

The R_SRVGAIN register is a servo loop gain setting register.
It adjusts the band and stability of the filter by changing the tracking loop filter coefficients (DC gain) from 75 to 400% in steps of 3.125%.

Gain calibration value = 100 (1+ R_SRVGAIN/32) [%]

Gain calibration value [%]	**R_SRVGAIN[7:0]**		**Default**
	HEX	DEC	
400	0x7F	127	
	-	-	
	0x61	97	
	0x60	96	
396.875	0x5F	95	
-	-	-	
106.250	0x02	2	
103.125	0x01	1	
100	0x00	0	●
96.875	0xFF	-1	
93.750	0xFE	-2	
-	-	-	
78.125	0xF9	-7	
75	0xF8	-8	
	0xF7	-9	
	-	-	
	0x80	-128	

The gain calibration value is calculated as the appropriate setting when performing angle INL error calibration.
For Shaft-End configurations without angle INL error calibration, please set this value to 0x00.

11-2. R_SRV

The R_SRV register is set to data when the angle INL error calibration of Off-Axis is performed, or when the anomaly magnetic field detection and error reduction functions are used.
Please set this register to 0x0 (Default) except for the above. When this setting is set to 0x1, the magnetic flux density data read out by the R_MAG register is not the magnetic field itself that is applied to the IC, but the data with a specific signal processing.

Conditions	**R_SRV**	**Default**
Normal Operation	0x0	●
When Off-Axis angle INL calibration is performed	0x1	
When using the function of anomaly magnetic field detection. When using the function to reduce errors due to anomaly magnetic fields.		

11-3. R_FILTER
The R_FILTERMODE register is a tracking loop filter bandwidth setting register.
A higher setting of this band improves the response of the angle output at acceleration and deceleration, but also increases the noise level.

Filter Bandwidth	**R_FILTER [2:0]**	**Default**
Single mode	0x0	
Narrow band width	0x1	●
Off-Axis angle INL calibration Mode	0x2	
AKM Reserved	0x3	
Double mode	0x4	
4-times mode	0x5	
8-times mode	0x6	
16-times mode	0x7	

Note19: Use 0x2 only when performing angle INL error calibration in off-axis configuration.
Do not use 0x2 in normal operation. Also, please do not use 0x3.

12) R_MLK (Addr. 0x0E)

(Addr.0x0E) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										R_MLK[1:0]	

The R_MLK register is an EEPROM write enable / disable indicator. To check whether or not writing to EEPROM is possible, read the R_MLK register using ReadReg OPCODE in User Mode.

EEPROM Condition	**R_MLK[1:0]**	**Default**
Locked	0x0	
Locked	0x1	
Locked	0x2	
Unlocked	0x3	●

The registers listed in items from 13 to 22 below are used for angle INL error calibration.

In this datasheet, contents concerning the calibration method are not described.
Please contact us for information of the calibration method.

In the default state, R_CA and R_GM have unique values for each sample, and 0x000 is written to the other registers.

13) R_PCAL1A (Addr. 0x0F)

(Addr.0x0F) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	R_PCAL1A[10:0]										

14) R_PCAL1P (Addr. 0x10)

(Addr.0x10) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_PCAL1P[11:0]											

15) R_PCAL2A (Addr. 0x11)

(Addr.0x11) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	R_PCAL2A[10:0]										

16) R_PCAL2P (Addr. 0x12)

(Addr.0x12) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_PCAL2P[11:0]											

17) R_PCAL3A (Addr. 0x13)

(Addr.0x13) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	R_PCAL3A[10:0]										

18) R_PCAL3P (Addr. 0x14)

(Addr.0x14) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_PCAL3P[11:0]											

19) R_PCAL4A (Addr. 0x15)

(Addr.0x15) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	R_PCAL4A[10:0]										

20) R_PCAL4P (Addr. 0x16)

(Addr.0x16) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_PCAL4P[11:0]											

21) R_CA (Addr. 0x17)

(Addr.0x17) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_CA[11:0]											

22) R_GM (Addr. 0x18)

(Addr.0x18) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_GM[11:0]											

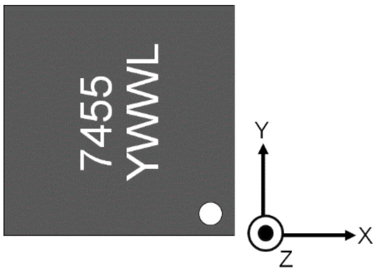
23) R_FSMF (Addr. 0x19)

(Addr.0x19) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
R_FIELDSEL [1:0]		R_CAL_RANGE [2:0]			R_MAG CAL	R_BACKCAL [1:0]		R_CAL_LMT[3:0]			

23-1. R_FIELDSEL

The R_FIELDSEL register selects the direction of the magnetic field to be detected. By changing the direction of the magnetic field, it is possible to change the arrangement of the IC and the magnet.

Direction of detection magnetic field	R_FIELDSEL [1:0]	Default
XY axes direction	0x0	●
ZY axes direction	0x1	
XZ axes direction	0x2	
XY axes direction	0x3	



Definition of X, Y and Z directions

Note20:
The magnetic field in the Z-axis is detected from HE_X when the ZY axis direction is selected.
The magnetic field in the Z-axis is detected from HE_Y when the XZ axis direction is selected.

The addresses listed in paragraphs 23.2 to 32 are registers for anomalous magnetic field detection and error reduction.
This datasheet does not contain information on how to set up these registers.
Please contact us for more information on how to set up these registers.

23-2. R_CAL_RANGE

The R_CAL_RANGE register sets the threshold for detecting an anomaly magnetic field. If the angular error due to the anomaly magnetic field exceeds the following threshold value(typ), an error is issued.

Threshold of Angular Error	R_CAL_RANGE [2:0]	Default
±5.6°	0x0	●
±4.2°	0x1	
±2.8°	0x2	
±1.4°	0x3	
±0.7°	0x4	
±2.1°	0x5	
±3.5°	0x6	
±4.9°	0x7	

23-3. R_MAGCAL

The R_MAGCAL register holds the ON/OFF setting data of the anomaly magnetic field detection and error reduction functions.
Set this register to 0x1 when using the function for detecting an anomaly magnetic field or reducing errors due to an anomaly magnetic field.

Note21: To activate the magnetic field error detection function, furthermore, activate the setting of R_DIS_ERR[3].
The R_BACKCAL setting must also be enabled for the field error reduction functions.

Anomalous magnetic field detection and error reduction	R_MAGCAL	Default
OFF	0x0	●
ON	0x1	

23-4. R_BACKCAL

The R_ BACKCAL register sets the angular error components that are subject to the functions of detecting anomaly magnetic fields and reducing the errors caused by them. This setting is valid when R_MAGCAL is set to 1.
To use only the magnetic anomaly detection function, set this parameter to 0x0.
The setting of 0x0 enables the magnetic field detection function for 1st and 2nd harmonic component errors.
In the case of setting other than 0x0, the magnetic field anomaly detection and error reduction functions will be applied as shown in the table below.

Error component	R_BACKCAL [1:0]	Default
Error reduction function disabled	0x0	●
1st harmonic component	0x1	
2nd harmonic component	0x2	
1st and 2nd harmonic component	0x3	

23-5. R_CAL_LMT

R_CAL_LMT register sets the upper limit of the calibration amount per rotation for the magnetic field error reduction function. The larger the value, the faster the calibration can be made, but ringing may occur.

Maximum calibration amount (Note22)	**R_CAL_LMT [3:0]**	**Default**
No calibration	0x0	●
MAX1LSB / rotation	0x1	
MAX2LSB / rotation	0x2	
-	-	
MAX13LSB / rotation	0xD	
MAX14LSB / rotation	0xE	
No upper limit. Calibrates the calculation result as it is.	0xF	

Note22: This setting applies to the angle INL error calibration parameters.

The addresses listed in paragraphs 24-32 are registers for error reduction due to anomalous magnetic fields.

24) R_SIN1_INIT (Addr. 0x1A)

(Addr.0x1A) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_SIN1_INIT[8:0]								

25) R_COS1_INIT (Addr. 0x1B)

(Addr.0x1B) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_COS1_INIT[8:0]								

26) R_SIN2_INIT (Addr. 0x1C)

(Addr.0x1C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_SIN2_INIT[8:0]								

27) R_COS2_INIT (Addr. 0x1D)

(Addr.0x1D) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_COS2_INIT[8:0]								

28) R_SIN1 (Addr. 0x1E)

(Addr.0x1E) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_SIN1[8:0]								

29) R_COS1 (Addr. 0x1F)

(Addr.0x1F) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_COS1[8:0]								

30) R_SIN2 (Addr. 0x20)

(Addr.0x20) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_SIN2[8:0]								

31) R_COS2 (Addr. 0x21)

(Addr.0x21) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			R_COS2[8:0]								

32) R_CALRDY (Addr. 0x22)

(Addr.0x22) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-											R_CAL RDY

20. EEPROM Address Map / Description

➢ EEPROM Address Map

Address	EEPROM Name	Permission		Contents
		Normal Mode	User Mode	
0x04	E_ID1[11:0]	N/A	R/W	Free EEPROM for User
0x08	E_ID2[11:0]		R/W	Free EEPROM for User
0x0C	E_ZP_U[1:0]		R/W	14bit Zero-point setting data (Upper 2bit)
0x10	E_ZP_L[11:0]		R/W	14bit Zero-point setting data (Lower 12bit)
0x14	E_RDABZ[11:0]		R/W	Diagnostic function setting Rotation direction setting Z-phase pulse phase setting ABZ output ON/OFF setting ABZ output hysteresis setting
0x18	E_ZW_U[1:0]		R/W	Z-phase pulse width Setting (Upper 2bit)
0x1C	E_ZW_L[11:0]		R/W	Z-phase pulse width setting (Lower 12bit)
0x20	E_ABZ_RES[11:0]		R/W	ABZ output resolution setting
0x24	E_UVW[11:0]		R/W	UVW output ON/OFF Setting UVW output hysteresis setting UVW output resolution setting 14bit Zero-point setting For UVW (Upper 2bit)
0x28	E_ZP_UVW_L[11:0]		R/W	14bit Zero-point setting For UVW (Lower 12bit)
0x2C	E_SRV_SET[11:0]		R/W	Servo filter setting Angle INL error calibration data
0x30	E_MLK[1:0]		R/W	Memory lock setting
0x34	E_PCAL1A[10:0]		R/W	Angle INL error calibration data
0x38	E_PCAL1P[11:0]		R/W	
0x3C	E_PCAL2A[10:0]		R/W	
0x40	E_PCAL2P[11:0]		R/W	
0x44	E_PCAL3A[10:0]		R/W	
0x48	E_PCAL3P[11:0]		R/W	
0x4C	E_PCAL4A[10:0]		R/W	
0x50	E_PCAL4P[11:0]		R/W	
0x54	E_CA[11:0]		R/W	
0x58	E_GM[11:0]		R/W	
0x5C	E_FSMF[11:0]		R/W	Direction of magnetic field detection, Anomaly magnetic field detection and error reduction settings
0x60	E_SIN1_INIT[8:0]		R/W	Anomaly magnetic field detection and error reduction settings
0x64	E_COS1_INIT[8:0]		R/W	
0x68	E_SIN2_INIT[8:0]		R/W	
0x6C	E_COS2_INIT[8:0]		R/W	

(N/A: not available, R: read-only, W: write-only, R/W: full access)

➢ EEPROM description

1) E_ID1, E_ID2 (Addr.0x04, 0x08)

(Addr.0x04) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_ID1[11:0]											

(Addr.0x08) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_ID2[11:0]											

E_ID1 and E_ID2 EEPROM are the memory that can freely write data (12bit) such as solid identification information or lot information. The default value is 0x000.

2) E_ZP (Addr.0x0C, 0x10)

(Addr.0x0C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										E_ZP_U[1:0]	

(Addr.0x10) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_ZP_L[11:0]											

The E_ZP EEPROM contains the 14bit configuration data of zero-degree point. This EEPROM is used to set an arbitrary angle position to zero (angle output 0º).
To validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_ZP.

Zero Point Position[º]	E_ZP_U[1:0]	E_ZP_L[11:0]	Default
0	0x0	0x000	●
0.022	0x0	0x001	
0.044	0x0	0x002	
0.066	0x0	0x003	
-	-	-	
89.978	0x0	0xFFF	
90.000	0x1	0x000	
-	-	-	
359.956	0x3	0xFFE	
359.978	0x3	0xFFF	

Note23: The value set in this EEPROM is reflected in SPI absolute angle output and ABZ output. The zero-degree point position for the UVW phase is set by E_ZP_UVW.

3) E_RDABZ (Addr. 0x14)

(Addr.0x14) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_DIS_ERR [3:0]				E_RD	E_Z_PHASE [1:0]		E_ABZ_E	E_ABZ_HYS[3:0]			

The E_RDABZ EEPROM contains enable/disable setting data of diagnosis function, rotation direction setting data, Z-phase pulse phase setting data, ABZ phase output ON/OFF setting data, and ABZ phase output hysteresis setting data.
To validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_RDABZ.

3-1. E_DIS_ERR
The E_DIS_ERR EEPROM stores ON/OFF setting data of each self-diagnosis function. To turn off the self-diagnosis function, set the corresponding bit to “1”.

E_DIS_ERR [3:0]	**Diagnosis**	**Default**
E_DIS_ERR[3]	Anomaly magnetic field detection Applies to the detection of R_ERRMON[5:3]	0
E_DIS_ERR[2]	*AKM reserved*	0
E_DIS_ERR[1]	Low magnetic flux density detection Applies to the detection of R_ERRMON[1]	0
E_DIS_ERR[0]	Tracking lost state detection Applies to the detection of R_ERRMON[0]	0

Note24: Please set E_DIS_ERR[2] to 0. To enable the setting of E_DIS_ERR[3], in addition to the above, it is necessary to set E_MAGCAL=1.

3-2. E_RD
The E_RD EEPROM stores the configuration data of the rotation direction data. It defines the increase or decrease of the output angle data relative to the magnetic field rotation direction.

Even if the rotation position of the magnet is the same in the Shaft-End and the Off-Axis configuration, the angle output by the AK7455 differs because the direction of the magnetic field input to the AK7455 is different.

[Shaft-End Configuration]
When selecting CCW (counter-clockwise), the output angle data increases in response to counter-clockwise magnet rotation. Counter-clockwise is defined by the 1-6-7-12-13-18-19-24 pin order direction in a top view of the QFN-24 package.
When selecting CW (clockwise), the output angle data increases in response to clockwise magnet rotation. Clockwise is defined by the 24-19-18-13-12-7-6-1 pin order direction in a top view of the QFN-24 package.

Rotation Direction	**E_RD**	**Default**
CCW	0x0	●
CW	0x1	

[Off-Axis Configuration]
In the case of E_RD=0 (CCW), E_ZP_U[1:0]=0x0, E_ZP_L[11:0]=0x0, the direction of the magnetic field and the angle output value for each arrangement of the AK7455 at each magnet rotation angle are shown in Figure12.
Figure13 shows the relationship between the magnet rotation angle and the angle output of AK7455 in each arrangement shown in Figure12.

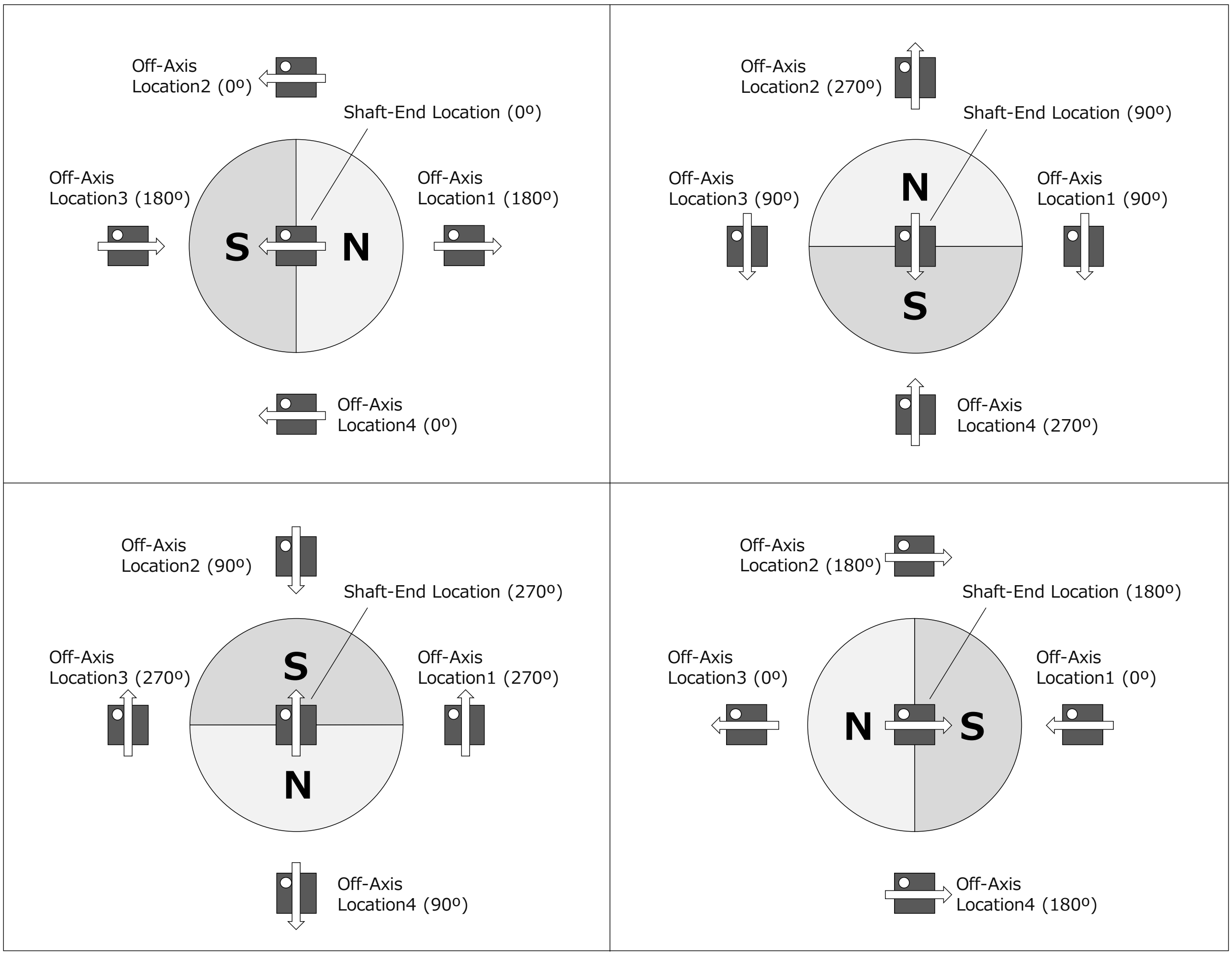


Figure12: Relationship between magnet position and angle in Shaft-End and Off-Axis

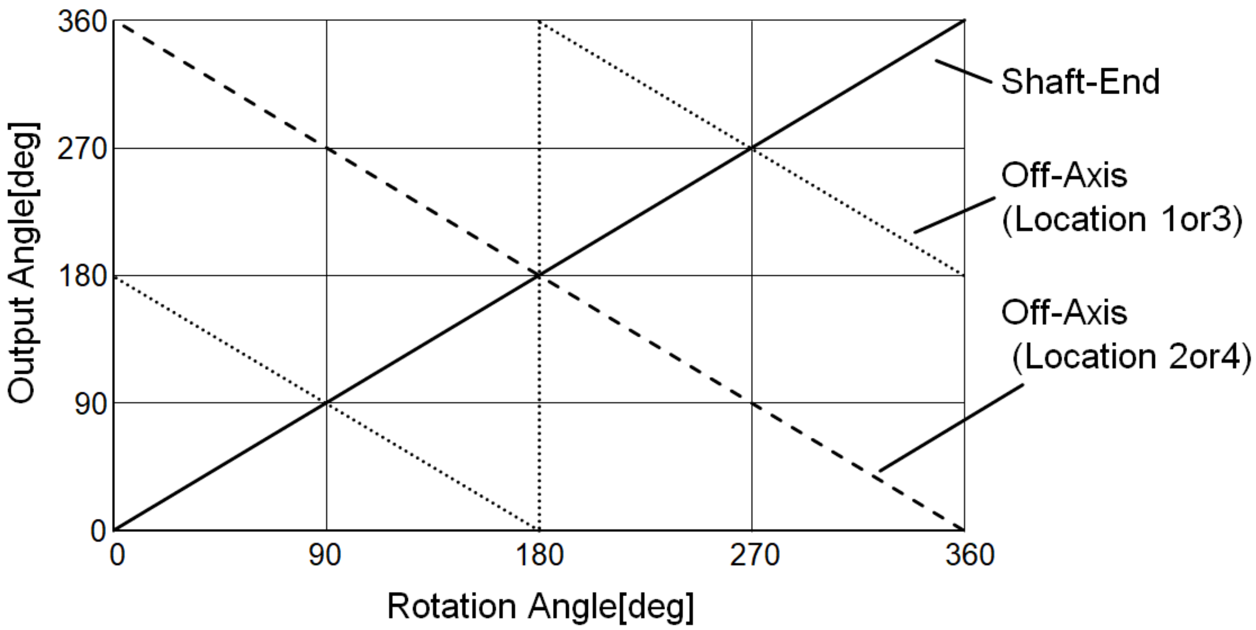


Figure13: Relationship between magnet rotation angle and output angle in different arrangements

3-3. E_Z_PHASE

The E_Z_PHASE EEPROM is the Z-phase pulse phase setting data. The rising edge of the Z-phase pulse can be synchronized with the following four types of edges.
Regardless of the setting, the Z-phase pulse rising edge is generated at an absolute angle of 0º obtained by SPI communication.

Z-phase Pulse Rising Edge	E_Z_PHASE[1:0]	Default
A-phase Pulse Falling Edge	0x0	●
B-phase Pulse Rising Edge	0x1	
A-phase Pulse Rising Edge	0x2	
B-phase Pulse Falling Edge	0x3	

Figure14 shows the relationship between the E_Z_PHASE setting value and the Z-phase pulse when E_RD=0, E_ZW_U=0, E_ZW_L=0, and the magnetic field rotation direction is CCW.

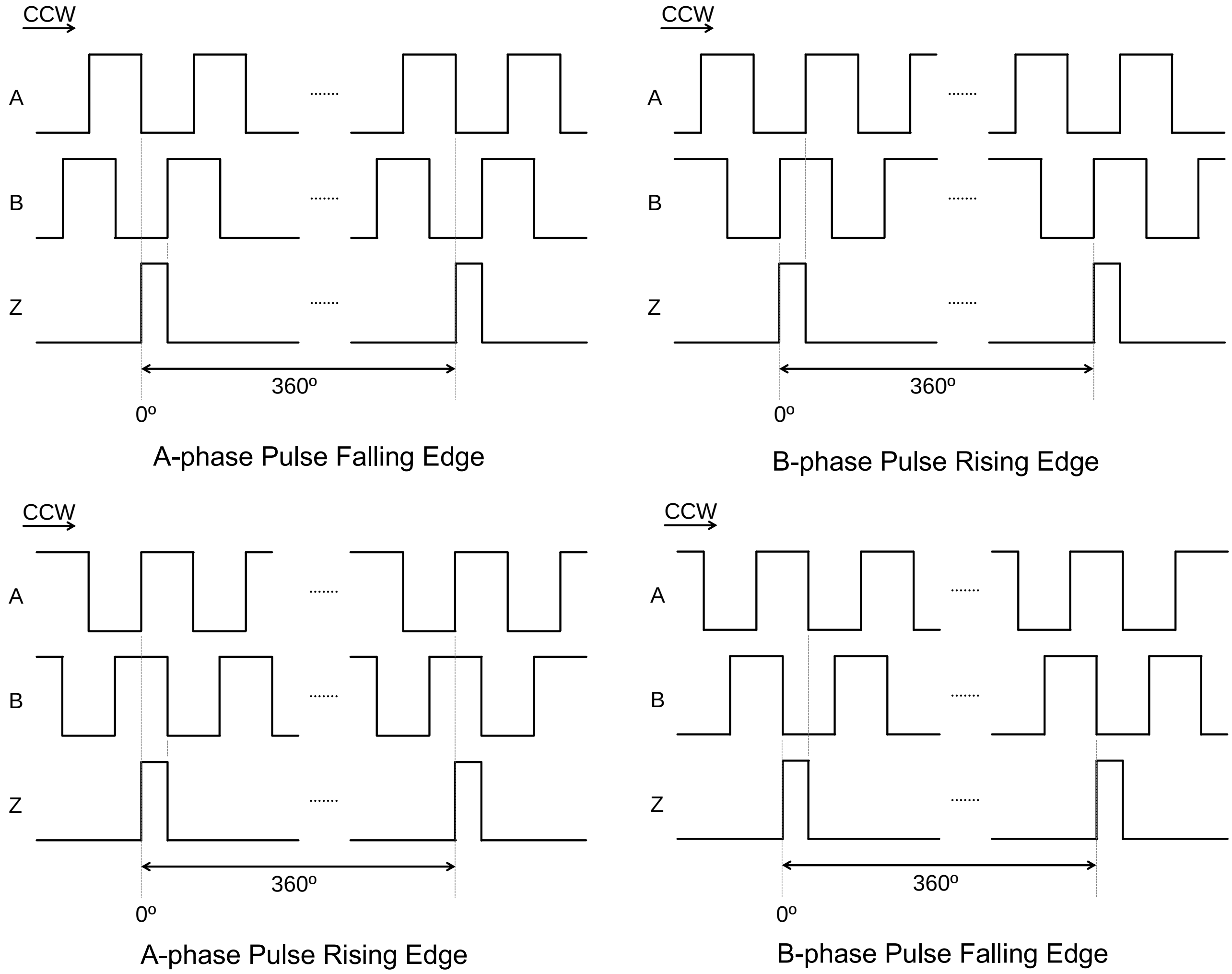


Figure14: Z-phase Pulse Phase Setting

3-4. E_ABZ_E

The E_ABZ_E EEPROM contains the configuration data of ABZ incremental output ON/OFF.
When selecting “OFF” setting, the A, B, and Z pins go to high-impedance (Hi-Z) output.

WARNING: During the start-up time after power-on and the transition from User Mode to Normal Mode (max: 40ms), the A, B and Z pins will be "L" regardless of this setting.

ABZ Output	E_ABZ_E	Default
OFF	0x0	
ON	0x1	●

3-5. E_ABZ_HYS

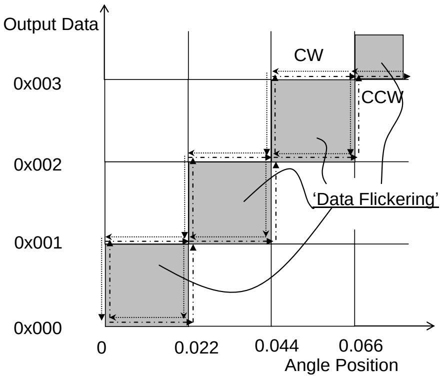
The E_ABZ_HYS EEPROM stores the configuration data of the hysteresis filter parameter.
This parameter defines whether the output angle data is updated or not in response to the processing of the raw angle data change. The output angle data is not updated when the processing of the raw angle data change is smaller than the hysteresis width. This parameter is defined in units of 1 LSB of 14bit angle data.
When the hysteresis width is increased, the data oscillation influence caused by a noisy environment (ex. module vibration) is decreased. However, the maximum output delay is increased as described below equation.

$$\text{Maximum Output Delay } T_D = 1.2 + 0.125(1+\text{“ABZ Hysteresis”}) \ [\mu s]$$

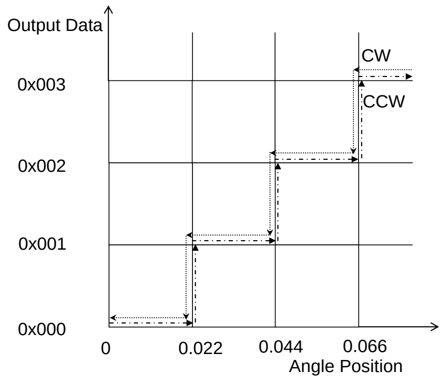
Note25: “ABZ Hysteresis” in the above equation is the number of LSB set by E_ABZ_HYS[3:0].
(Example: “ABZ Hysteresis" = 3 when E_ABZ_HYS[3:0]=0x4.)

When selecting the “OFF” setting, the output angle data equals the processing of the raw angle data. The inherent LSB data flickering (Note26) is not eliminated from the output angle data. The maximum output delay is the shortest (1.2μs) in this setting.
When selecting the “0LSB” setting, the inherent LSB data flickering is eliminated from the output angle data. The maximum output delay is 1.325μs in this setting.

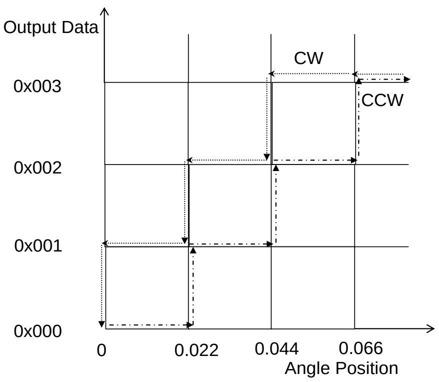
Note26: The LSB data flickering inheres in the type2 tracking loop architecture. This data flickering appears even in the ideal environment under a noise-free case. Therefore, AKM does not recommend the “OFF” setting.



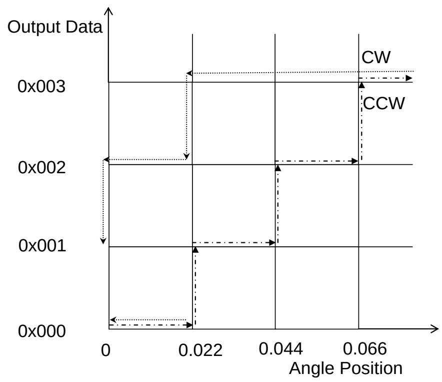
“OFF” setting



“0LSB” setting



“1LSB” setting



“2LSB” setting

Figure15: ABZ Hysteresis Setting

The relationship between E_ABZ_HYS[3:0] and the hysteresis set value is as follows.

ABZ Hysteresis	E_ABZ_HYS[3:0]	Default
OFF	0x0	
0LSB	0x1	
1LSB	0x2	●
2LSB	0x3	
3LSB	0x4	
4LSB	0x5	
5LSB	0x6	
6LSB	0x7	
7LSB	0x8	
8LSB	0x9	
9LSB	0xA	
10LSB	0xB	
11LSB	0xC	
12LSB	0xD	
OFF	0xE	
OFF	0xF	

4) E_ZW (Addr. 0x18, 0x1C)

(Addr.0x18) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										E_ZW_U[1:0]	

(Addr.0x1C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_ZW_L[11:0]											

The E_ZW EEPROM contains the Z-phase pulse width setting data of the ABZ incremental output.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_ZW.
It can be set to any value from 1 to 16384LSB. 1LSB is 1/4 of the A phase 1 period set by E_ABZ_RES (ABZ phase resolution setting).
When the setting value satisfies the condition of E_ZW+1≧4(E_ABZ_RES+1), the Z-phase pulse always goes to the “H” level.

Z-phase Pulse Width[LSB]	E_ZW_U[1:0]	E_ZW_L[11:0]	Default
1	0x0	0x000	
2	0x0	0x001	
3	0x0	0x002	
4	0x0	0x003	●
-	-	-	
4096	0x0	0xFFF	
4097	0x1	0x000	
-	-	-	
16383	0x3	0xFFE	
16384	0x3	0xFFF	

Figure16 shows an example of the Z-phase pulse width when E_RD=0x0, E_ABZ_RES=0x9C3 (Default) and the magnetic field rotation direction is CCW.

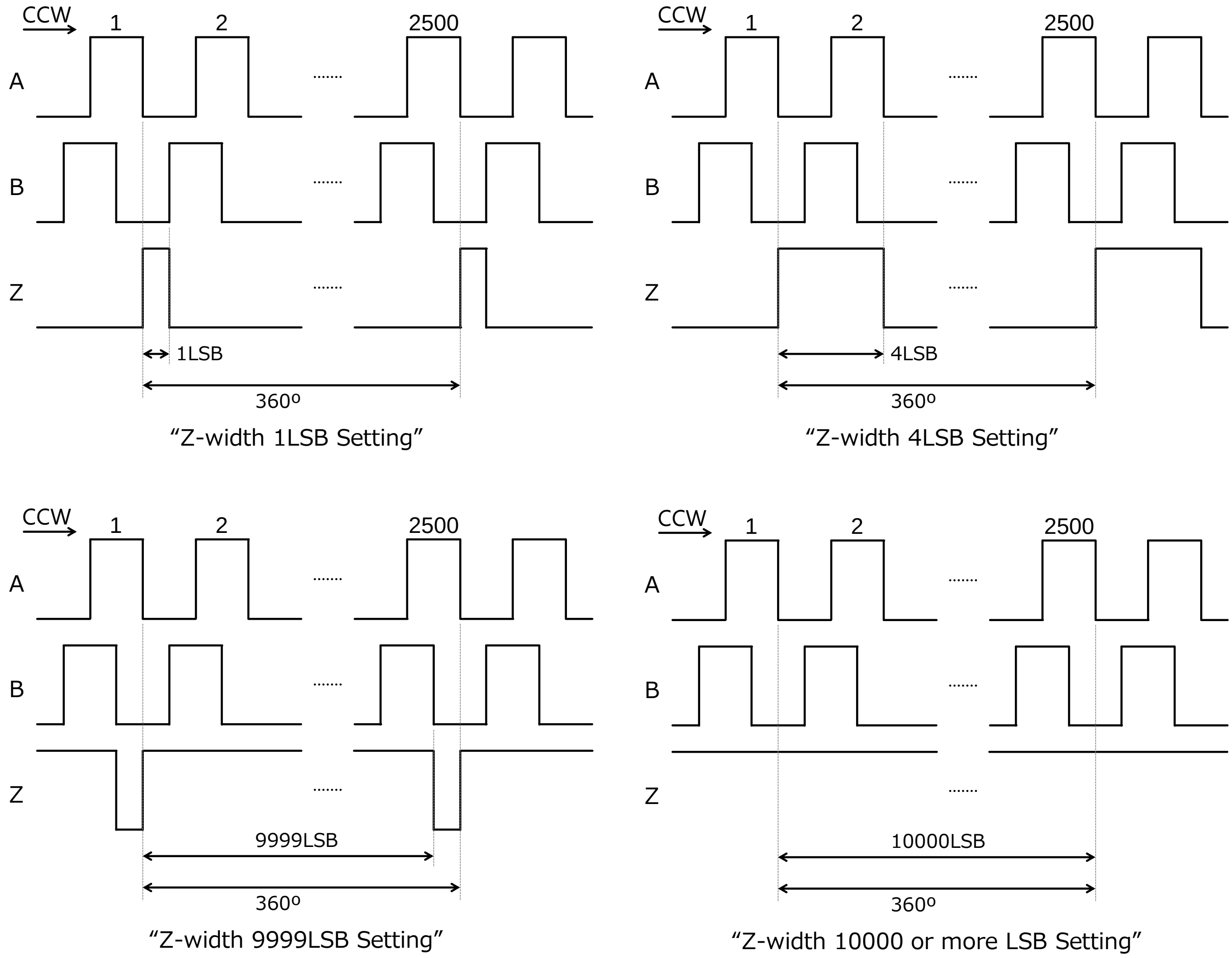


Figure16: Z-phase Pulse Width Setting

5) E_ABZ_RES (Addr. 0x20)

<table>
<tr><td colspan="12">(Addr.0x20) DATA[11:0]</td></tr>
<tr><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="12">E_ABZ_RES[11:0]</td></tr>
</table>

The E_ABZ_RES EEPROM contains the configuration data of ABZ incremental output resolution.
It can be set to an arbitrary value from 1 to 4096ppr.

ABZ Resolution	E_ABZ_RES[11:0]	Default
1ppr	0x000	
2ppr	0x001	
-	-	
1000ppr	0x3E7	
-	-	
2500ppr	0x9C3	●
-	-	
4095ppr	0xFFE	
4096ppr	0xFFF	

In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_ABZ_RES.

6) E_UVW (Addr. 0x24)

<table>
<tr><td colspan="12">(Addr.0x24) DATA[11:0]</td></tr>
<tr><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td>E_UVW_E</td><td colspan="4">E_UVW_HYS[3:0]</td><td colspan="5">E_UVW_RES[4:0]</td><td colspan="2">E_ZP_UVW_U[1:0]</td></tr>
</table>

The E_UVW contains the configuration data of UVW output ON/OFF, UVW output hysteresis, UVW output resolution and the UVW phase 14-bit zero-point data (upper 2 bits).
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_UVW.

6-1. E_UVW_E
The E_UVW_E EEPROM stores the configuration data of the UVW commutation output ON/OFF.
When selecting the “OFF” setting, the U, V, and W pins go to high-impedance (Hi-Z) output.

WARNING: During the start-up time after power-on and the transition from User Mode to Normal Mode (max: 40ms), the U, V, and W pins will be "L" regardless of this setting.

UVW Output	E_UVW_E	Default
OFF	0x0	
ON	0x1	●

6-2. E_UVW_HYS
The E_UVW_HYS EEPROM stores the configuration data of the hysteresis filter parameter.
This parameter defines whether the UVW commutation output is updated or not in response to the processing raw angle data change. The UVW commutation output is not updated when the processing of the raw angle data change is smaller than the hysteresis parameter. This parameter is defined in units of 1LSB of 14bit angle data.

WARNING: This hysteresis parameter works independently of the E_ABZ_HYS setting.

The relationship between E_UVW_HYS[3:0] and the hysteresis setting value is as follows.

UVW Hysteresis	E_UVW_HYS[3:0]	Default
OFF	0x0	
0LSB	0x1	
1LSB	0x2	●
2LSB	0x3	
3LSB	0x4	
4LSB	0x5	
5LSB	0x6	
6LSB	0x7	
7LSB	0x8	
8LSB	0x9	
9LSB	0xA	
10LSB	0xB	
11LSB	0xC	
12LSB	0xD	
OFF	0xE	
OFF	0xF	

6-3. E_UVW_RES

The E_UVW_RES EEPROM stores the configuration data of the UVW commutation output resolution. It can be set to an arbitrary value from 2 to 64poles (1 to 32pole pairs).

UVW Resolution	E_UVW_RES[4:0]	Default
2 poles (1 pole pair)	0x00	
4 poles (2 pole pairs)	0x01	
6 poles (3 pole pairs)	0x02	
8 poles (4 pole pairs)	0x03	●
10 poles (5 pole pairs)	0x04	
12 poles (6 pole pairs)	0x05	
14 poles (7 pole pairs)	0x06	
16 poles (8 pole pairs)	0x07	
18 poles (9 pole pairs)	0x08	
20 poles (10 pole pairs)	0x09	
22 poles (11 pole pairs)	0x0A	
24 poles (12 pole pairs)	0x0B	
26 poles (13 pole pairs)	0x0C	
28 poles (14 pole pairs)	0x0D	
30 poles (15 pole pairs)	0x0E	
32 poles (16 pole pairs)	0x0F	
34 poles (17 pole pairs)	0x10	
36 poles (18 pole pairs)	0x11	
38 poles (19 pole pairs)	0x12	
40 poles (20 pole pairs)	0x13	
42 poles (21 pole pairs)	0x14	
44 poles (22 pole pairs)	0x15	
46 poles (23 pole pairs)	0x16	
48 poles (24 pole pairs)	0x17	
50 poles (25 pole pairs)	0x18	
52 poles (26 pole pairs)	0x19	
54 poles (27 pole pairs)	0x1A	
56 poles (28 pole pairs)	0x1B	

58 poles (29 pole pairs)	0x1C	
60 poles (30 pole pairs)	0x1D	
62 poles (31 pole pairs)	0x1E	
64 poles (32 pole pairs)	0x1F	

7) E_ZP_UVW_L (Addr. 0x28)

(Addr.0x28) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_ZP_UVW_L[11:0]											

The lower 2 bits of the Addr. 0x24: E_ZP_UVW_U[1:0] and this EEPROM contain 14-bit zero point information data.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_ZP_UVW_L.

These are EEPROMs to set the arbitrary angle position to UVW phase zero point (angle output 0º). It operates independently of the zero-point setting (E_ZP) for the ABZ phase.

Zero Point Position[º]	E_ZP_UVW_U[1:0]	E_ZP_UVW_L[11:0]	Default
0	0x0	0x000	●
0.022	0x0	0x001	
0.044	0x0	0x002	
0.066	0x0	0x003	
-	-	-	
89.978	0x0	0xFFF	
90.000	0x1	0x000	
-	-	-	
359.956	0x3	0xFFE	
359.978	0x3	0xFFF	

Note27: The above zero position is set with respect to the absolute angle position when E_ZP=0x0 is set.

8) E_SRV_SET (Addr. 0x2C)

(Addr.0x2C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_SRVGAIN[7:0]								E_SRV	E_FILTER[2:0]		

8-1. E_SRVGAIN

The E_SRVGAIN is a servo loop gain setting EEPROM.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_SRVGAIN.

It adjusts the band and stability of the filter by changing the tracking loop filter coefficients (DC gain) from 75 to 400% in steps of 3.125%.

Gain calibration value =100 (1 + E_SRVGAIN/32) [%]

Gain calibration value [%]	E_SRVGAIN[7:0]		Default
	HEX	DEC	
400	0x7F	127	
	-	-	
	0x61	97	
	0x60	96	
396.875	0x5F	95	
-	-	-	
103.125	0x01	1	
100	0x00	0	●
96.875	0xFF	-1	
93.75	0xFE	-2	
-	-	-	
75	0xF8	-8	
	0xF7	-9	
	-	-	
	0x80	-128	

The gain calibration value is calculated as the appropriate setting when performing angle INL error calibration.
For Shaft-End configurations without angle INL error calibration, please set this value to 0x00.

8-2. E_SRV

The E_SRV is set to data when the angle INL error calibration of Off-Axis is performed, or when the anomaly magnetic field detection and error reduction functions are used.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_SRV.

Please set this EEPROM to 0x0 (Default) except for the above. When this setting is set to 0x1, the magnetic flux density data read out by the R_MAG register is not the magnetic field itself that is applied to the IC, but the data with a specific signal processing.

Conditions	**E_SRV**	**Default**
Normal Operation	0x0	●
When Off-Axis angle INL calibration is performed	0x1	
When using the function of anomaly magnetic field detection. When using the function to reduce errors due to anomaly magnetic fields.		

8-3. E_FILTER

The E_FILTER is a tracking loop filter bandwidth setting EEPROM.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_FILTER.

A higher setting of this band improves the response of the angle output at acceleration and deceleration, but also increases the noise level.

Filter Bandwidth	**E_FILTER [2:0]**	**Default**
Single mode	0x0	
Narrow band width	0x1	●
Off-Axis Angle INL calibration Mode	0x2	
AKM Reserved	0x3	
Double mode	0x4	
4-times mode	0x5	
8-times mode	0x6	
16-times mode	0x7	

Note28: Use 0x2 only when performing angle INL error calibration in off-axis configuration.
Do not use 0x2 in normal operation. Also, please do not use 0x3.

9) E_MLK (Addr. 0x30)

(Addr.0x30) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-										E_MLK[1:0]	

The E_MLK EEPROM contains writing lock setting data. To prevent erroneous writing to the EEPROM, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_MLK.

Memory Condition	E_MLK[1:0]	Default
Locked	0x0	
Locked	0x1	
Locked	0x2	
Unlocked	0x3	●

WARNING: Once a value other than 0x3 is written to E_MLK[1:0], all EEPROM including E_MLK[1:0] itself cannot be rewritten.

The EEPROMs listed in items from 10 to 19 below are used for angle INL error calibration.

In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in each EEPROM.

In this datasheet, contents concerning calibration method are not described.
Please contact us for information of calibration method.

In the default state, E_CA and E_GM have unique values for each sample, and 0x000 is written to the other EEPROMs.

10) E_PCAL1A (Addr. 0x34)

(Addr.0x34) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	E_PCAL1A[10:0]										

11) E_PCAL1P (Addr. 0x38)

(Addr.0x38) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_PCAL1P[11:0]											

12) E_PCAL2A (Addr. 0x3C)

(Addr.0x3C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	E_PCAL2A[10:0]										

13) E_PCAL2P (Addr. 0x40)

(Addr.0x40) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_PCAL2P[11:0]											

14) E_PCAL3A (Addr. 0x44)

(Addr.0x44) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	E_PCAL3A[10:0]										

15) E_PCAL3P (Addr. 0x48)

(Addr.0x48) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_PCAL3P[11:0]											

16) E_PCAL4A (Addr. 0x4C)

(Addr.0x4C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-	E_PCAL4A[10:0]										

17) E_PCAL4P (Addr. 0x50)

(Addr.0x50) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_PCAL4P[11:0]											

18) E_CA (Addr. 0x54)

(Addr.0x54) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_CA[11:0]											

19) E_GM (Addr. 0x58)

(Addr.0x58) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_GM[11:0]											

20) E_FSMF (Addr. 0x5C)

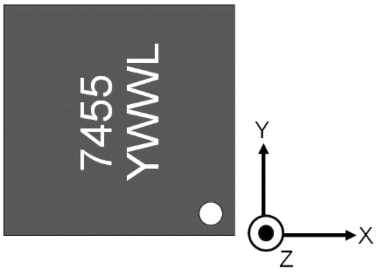
(Addr.0x5C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
E_FIELDSEL [1:0]		E_CAL_RANGE [2:0]			E_MAG CAL	E_BACKCAL [1:0]		E_CAL_LMT[3:0]			

E_FSMF EEPROM contains the setting data for magnetic field direction setting, anomalous field detection and error reduction.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in E_FSMF.

20-1. E_FIELDSEL

The E_FIELDSEL selects the direction of the magnetic field to be detected. By changing the direction of the magnetic field, it is possible to change the arrangement of the IC and the magnet.

Direction of detection magnetic field	E_FIELDSEL [1:0]	Default
XY axes direction	0x0	●
ZY axes direction	0x1	
XZ axes direction	0x2	
XY axes direction	0x3	



Definition of X, Y and Z directions

Note29:
The magnetic field in the Z-axis is detected from HE_X when the ZY axis direction is selected.
The magnetic field in the Z-axis is detected from HE_Y when the XZ axis direction is selected.

The addresses listed in paragraphs 20.2 to 24 are EEPROM for anomalous magnetic field detection and error reduction.
In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in each EEPROM.

This datasheet does not contain information on how to set up these EEPROMs.
Please contact us for more information on how to set up these EEPROMs.

20-2. E_CAL_RANGE

The E_CAL_RANGE sets the threshold for detecting an anomaly magnetic field. If the angular error due to the anomaly magnetic field exceeds the following threshold value(typ), an error is issued.

Threshold of Angular Error	E_CAL_RANGE [2:0]	Default
±5.6°	0x0	●
±4.2°	0x1	
±2.8°	0x2	
±1.4°	0x3	
±0.7°	0x4	
±2.1°	0x5	
±3.5°	0x6	
±4.9°	0x7	

20-3. E_MAGCAL

The E_MAGCAL holds the ON/OFF setting data of the anomaly magnetic field detection and error reduction functions.
Set this EEPROM to 0x1 when using the function for detecting an anomaly magnetic field or reducing errors due to an anomaly magnetic field.

Note30: To activate the magnetic field error detection function, furthermore, activate the setting of E_DIS_ERR[3].
The E_BACKCAL setting must also be enabled for the field error reduction functions.

Anomalous magnetic field detection and error reduction	E_MAGCAL	Default
OFF	0x0	●
ON	0x1	

20-4. E_BACKCAL

The E_ BACKCAL sets the angular error components that are subject to the functions of detecting anomaly magnetic fields and reducing the errors caused by them. This setting is valid when E_MAGCAL is set to 1.
To use only the magnetic anomaly detection function, set this parameter to 0x0.
The setting of 0x0 enables the magnetic field detection function for 1st and 2nd harmonic component errors.
In the case of setting other than 0x0, the magnetic field anomaly detection and error reduction functions will be applied as shown in the table below.

Error component	E_BACKCAL [1:0]	Default
Error reduction function disabled	0x0	●
1st harmonic component	0x1	
2nd harmonic component	0x2	
1st and 2nd harmonic component	0x3	

20-5. E_CAL_LMT

E_CAL_LMT sets the upper limit of the calibration amount per rotation for the magnetic field error reduction function. The larger the value, the faster the calibration can be made, but ringing may occur.

Maximum calibration amount (Note31)	E_CAL_LMT [3:0]	Default
No calibration	0x0	●
MAX1LSB / rotation	0x1	
MAX2LSB / rotation	0x2	
-	-	
MAX13LSB / rotation	0xD	
MAX14LSB / rotation	0xE	
No upper limit. Calibrates the calculation result as it is.	0xF	

Note31: This setting applies to the angle INL error calibration parameters.

The addresses listed in paragraphs 21-24 are EEPROMs for error reduction due to anomalous magnetic fields.

In order to validate the setting in Normal Mode, use the WriteEEPROM OPCODE in User Mode and store the setting data in each EEPROM.

21) E_SIN1_INIT (Addr. 0x60)

(Addr.0x60) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			E_SIN1_INIT[8:0]								

22) E_COS1_INIT (Addr. 0x64)

(Addr.0x64) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			E_COS1_INIT[8:0]								

23) E_SIN2_INIT (Addr. 0x68)

(Addr.0x68) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			E_SIN2_INIT[8:0]								

24) E_COS2_INIT (Addr. 0x6C)

(Addr.0x6C) DATA[11:0]											
11	10	9	8	7	6	5	4	3	2	1	0
-			E_COS2_INIT[8:0]								

21. Magnet angle position, output angle and rotation direction

Figure17 shows the output angle of the AK7455 at the time of shipment and the positional relationship between the package and the magnet (when the zero-point setting is the default value). Since these relationships have alignment errors of about a few degrees, if you want to accurately determine the positional relationship with the magnet, please adjust using the zero-point setting.

Also, the default value of the rotation direction setting is counterclockwise (CCW). CCW is defined as the direction of turning pin numbers 1-6-7-12-13-18-19-24 in the pin of QFN-24 package.

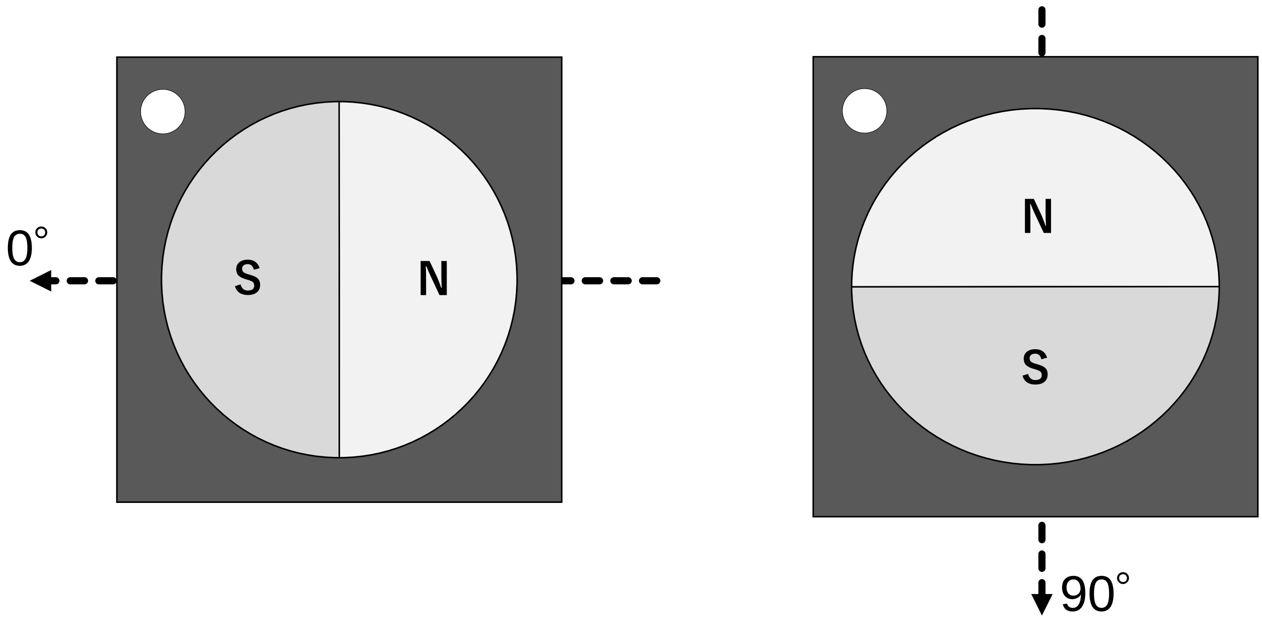
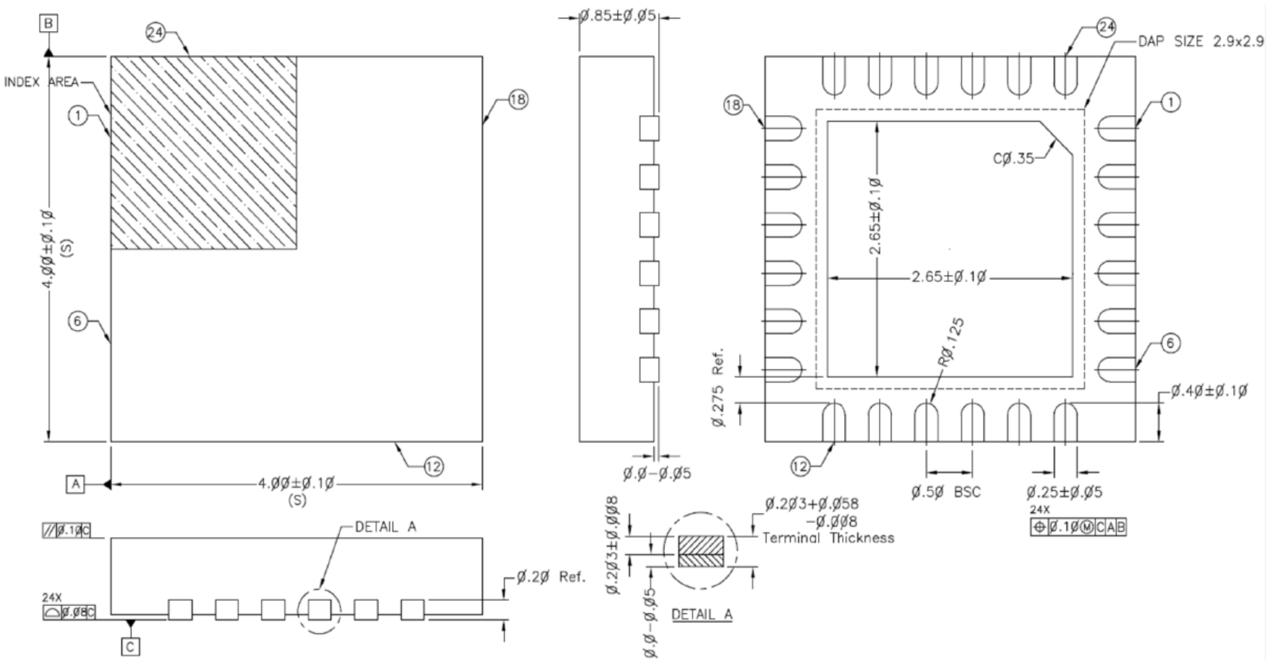


Figure17: Relationship between magnet position, output angle and rotation direction at factory default setting

22. Package Information

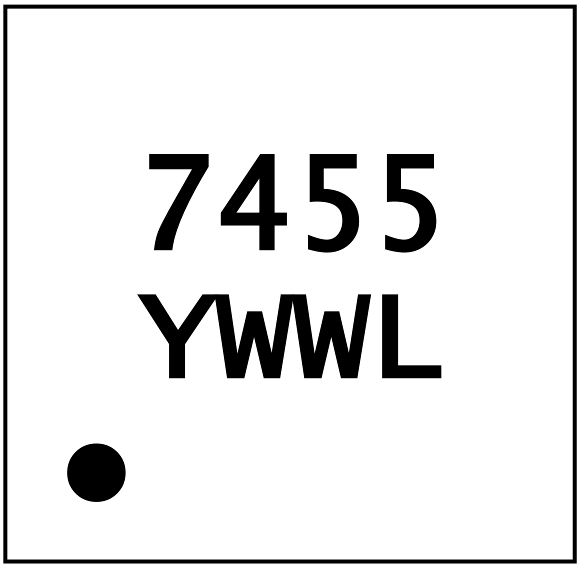
22.1. QFN24 – Package Dimensions



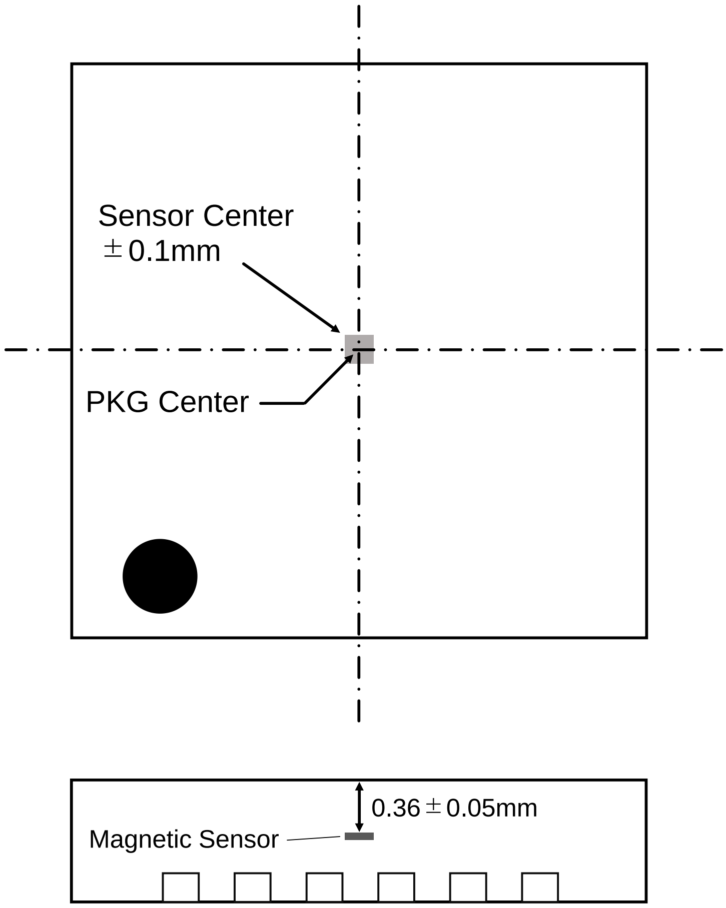
22.2. **Marking Information**

Product Name : 7455
Date code : YWWL

- Y = Year code
- WW = Week code
- L = Lot



22.3. Sensor Position Information



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